CHAPTER 3 Enabling Technologies

CONTENTS

- Accelerometer and Gyroscope
- Overview of Location Base Service
- Mobile communication
- GPS

Most materials are from books and online sources. For use in lectures only. Not for duplication.

Accelerometer and Gyroscope

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Enabling Technologies:Accelerometer

- An **accelerometer** measures acceleration

- Single- and multi-axis models can detect magnitude and direction of the acceleration:
  - orientation,
  - acceleration,
  - vibration shock, and
  - Falling



Nintendo's Wii

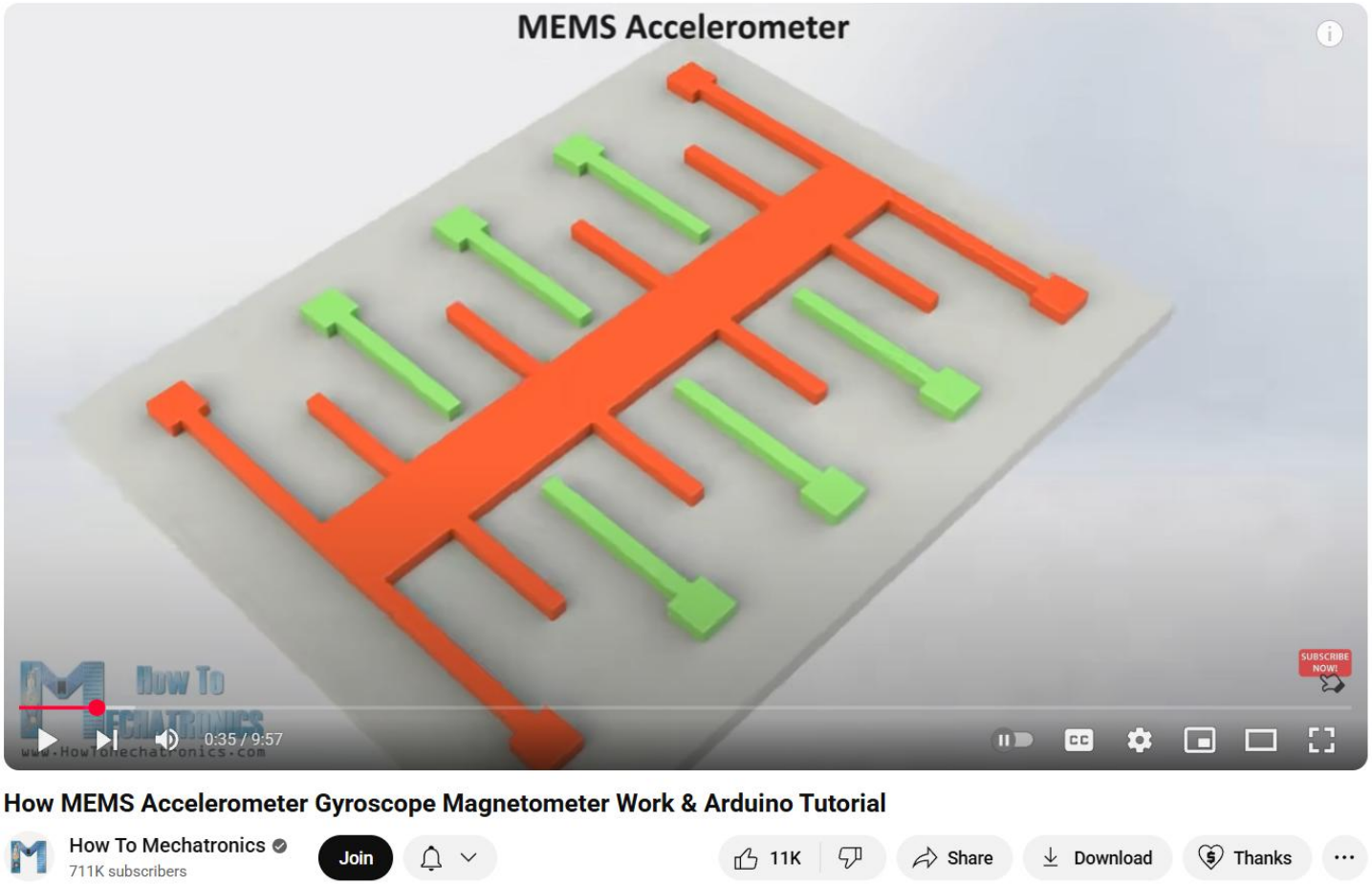
- Micro machined accelerometers are used in portable electronic devices and video game controllers for **positioning** or **input**

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Physical Principles

https://www.youtube.com/watch?v=eqZgxR6eRjo&t=34s



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

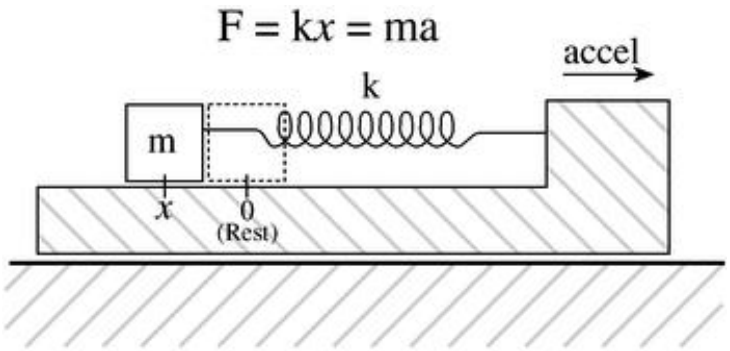
Physical principles

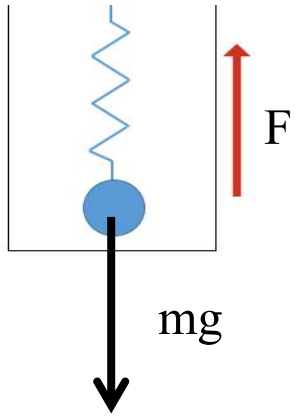
Example 3.1 Accelerometer

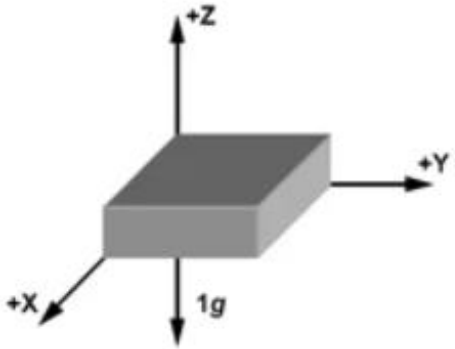
Q: What's the working principle of an accelerometer?

- An accelerometer acts like a damped mass on a spring.
- When it accelerates, the mass displaces, allowing the spring to accelerate it at the same rate as the casing.
- Displacement is measured to determine the acceleration using **Hooke's Law**
- Measures along **the three axes**, the **gravity has to be subtracted**

$$F - mg = kx = ma$$







Courtesy: Dr. Wilton Fok

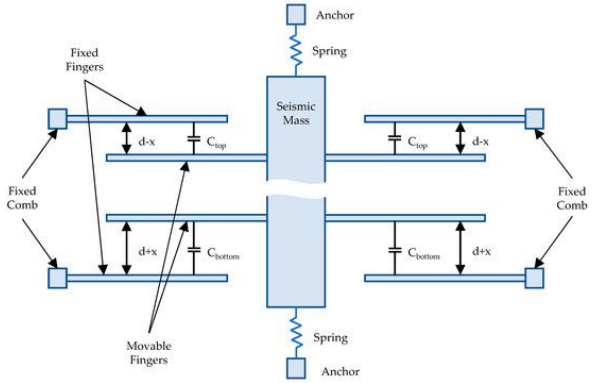
Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Physical principles

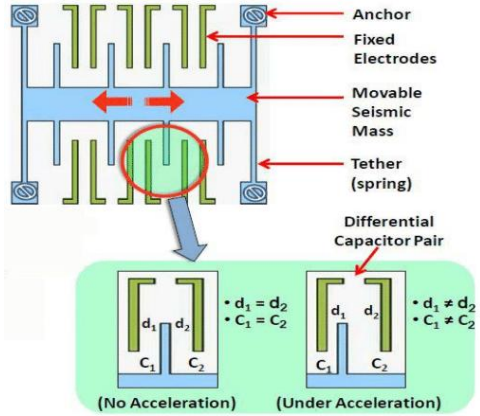
Example 3.2 Modern accelerometers

Q: Give examples of modern accelerometers.

- Modern accelerometers are often small *micro electro-mechanical systems* [1]
- Make use of **Piezoelectric Effect** to **convert mechanical stress to electric charges**.
- The design consists of two capacitors.
- When accelerated, the mass moves, and **one of the capacitances increases** while the other decreases.
- **The difference between these capacitances** can be measured to compute the acceleration



Vertical



Horizontal

[1]Szermer, M.; Zając, P.; Amrozik, P.; Maj, C.; Jankowski, M.; Jabłoński, G.; Kiełbik, R.; Nazdrowicz, J.; Napieralska, M.; Sakowicz, B. A Capacitive 3-Axis MEMS Accelerometer for Medipost: A Portable System Dedicated to Monitoring Imbalance Disorders. Sensors 2021, 21, 3564. https://doi.org/10.3390/s21103564

[2]Amerini, I., Bestagini, P., Bondi, L., Caldelli, R., Casini, M., & Tubaro, S. (2016). Robust smartphone fingerprint by mixing device sensors features for mobile strong authentication. In IS INTERNATIONAL SYMPOSIUM ON ELECTRONIC IMAGING (Vol. 2016, No. 8, pp. 1-8). Society for Imaging Science and Technology.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Navigation

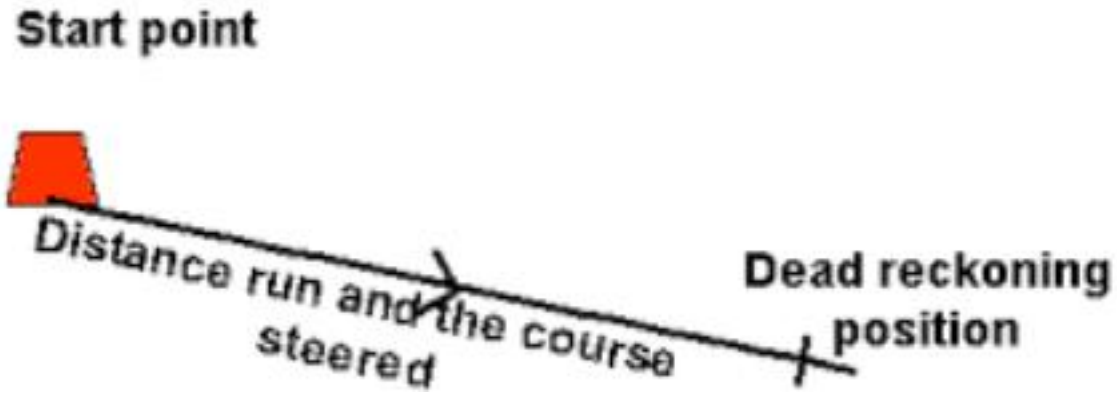
Example 3.3 Navigation

Q: What is an inertial naviation system?

GPS not used



- An **Inertial Navigation System** (INS) is a navigation aid that uses a computer and motion sensors (accelerometers) to continuously calculate via
  - **dead reckoning:**
    calculate orientation and velocity from last known position
  - without the need for external references (e.g. GPS)



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Applications

- Mobile phone/tablets use accelerometers to automatically align the screen
- i.e. switching between portrait and landscape modes.



Smartphones can download an **Automatic Collision Notification** (ACN) app

The phone's accelerometer **detects crash-strength G-forces** and automatically calls for assistance unless manually cancelled.



http://www.youtube.com/watch?v=KOwAPj3YiPw

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Applications (Cont.)

- **Nintendo's Wii** uses three-axis accelerometer and was designed primarily for motion input.



- **Camcorders** use accelerometers for image stabilization.
- **Still cameras** use accelerometers for anti-blur capturing.



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Device integrity

- Some laptops and mobile devices feature an accelerometer for **damage protection**
  - E.g.
    - Lenovo's (formerly IBM's) Active Protection System,
    - Apple's Sudden Motion Sensor
    - HP's 3D DriveGuard,



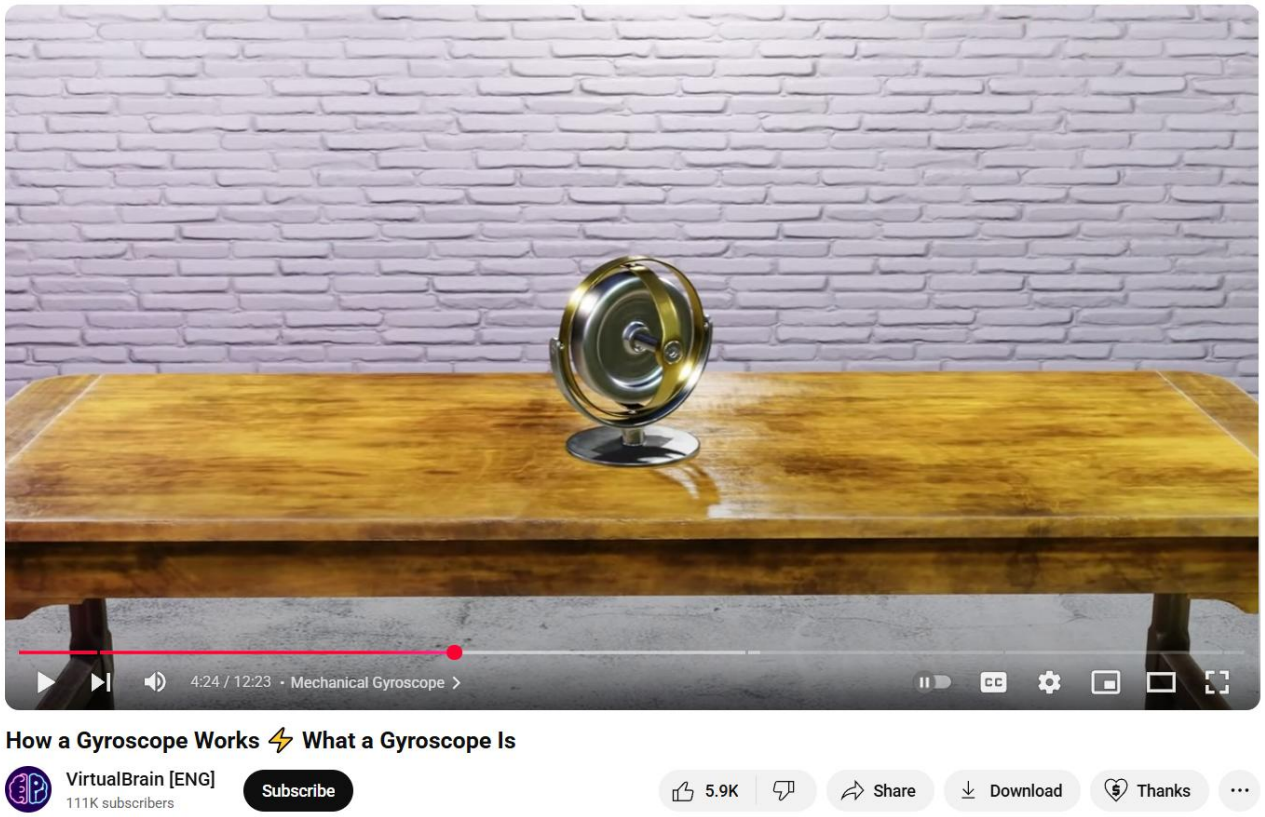
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Introduction to Gyroscope

How a Gyroscope works

https://youtu.be/V6XSsNAWg00?t=264



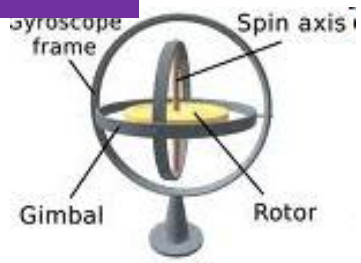
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Introduction to Gyroscope

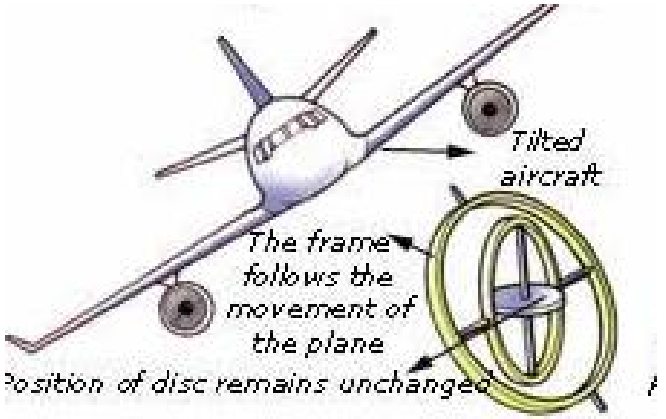
Example 3.4 Gyroscope

Q: Outline the working principle of a Gyroscope



http://en.wikipedia.org/wiki/Gyroscope

- **Spinning Motion:** A gyroscope consists of a wheel or rotor that spins rapidly around its axis, generating angular momentum..
- **Conservation of Angular Momentum:** The gyroscope resists changes to its rotational axis, maintaining its position even when external forces are applied.
- **Precession:** When an external torque (like gravity) acts on the tilted gyroscope, precession causes its axis to move in a circular path around the vertical.
- used in Navigation, stablization of UAVs, golf clubs

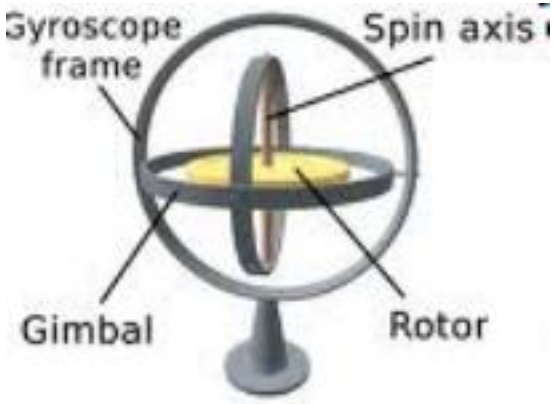


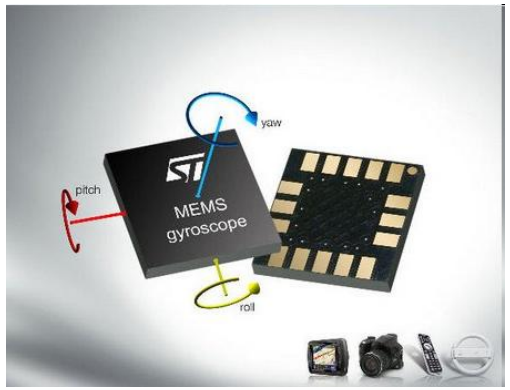


Courtesy: Dr. Wilton Fok

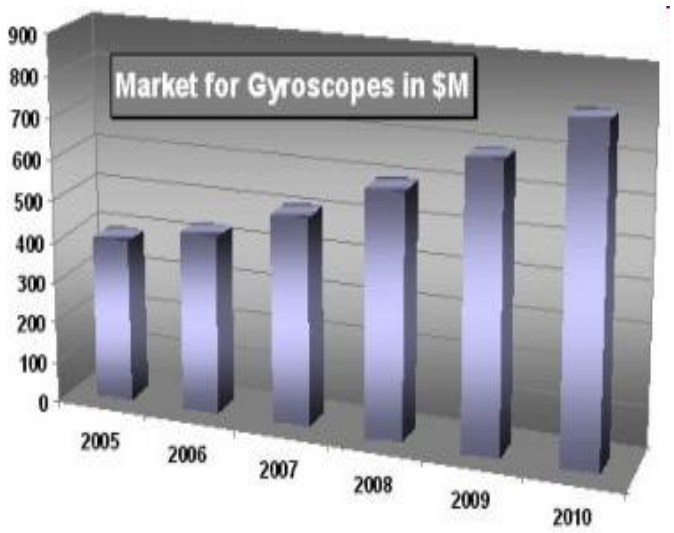
Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Digital gyroscope market forecast





- Digital gyroscope industry plans a high growth potential of the defence and low end automotive applications.



Digital Gyroscope Market is Expected to Reach 800M$ in 2010

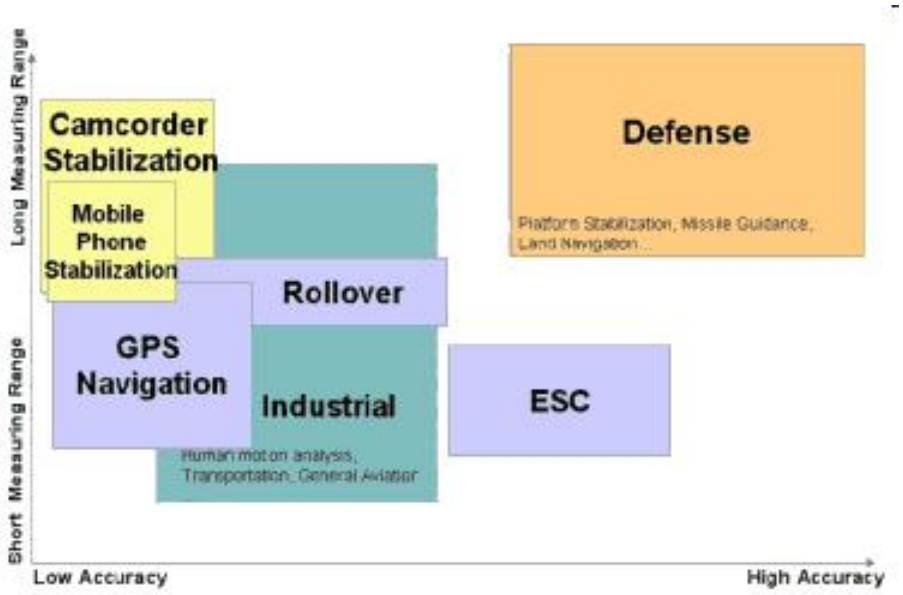
Sensors Portal. (2006, May). MEMS gyroscope market. http://www.sensorsportal.com/HTML/DIGEST/may_06/MEMS_Gyroscope_market.htm

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Digital gyroscope market forecast

- **3 fields with high growth potential**
  - **Automotive** applications -highest new entrant rate
  - **Defence & aeronautics** applications - stable business
  - **Mobile/Consumer** applications- booming markets thanks to sensor introduction in personal electronics). E.g. games



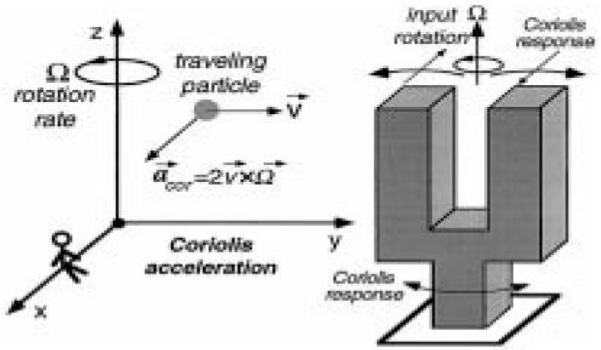
Sensors Portal. (2006, May). MEMS gyroscope market. http://www.sensorsportal.com/HTML/DIGEST/may_06/MEMS_Gyroscope_market.htm

Courtesy: Dr. Wilton Fok

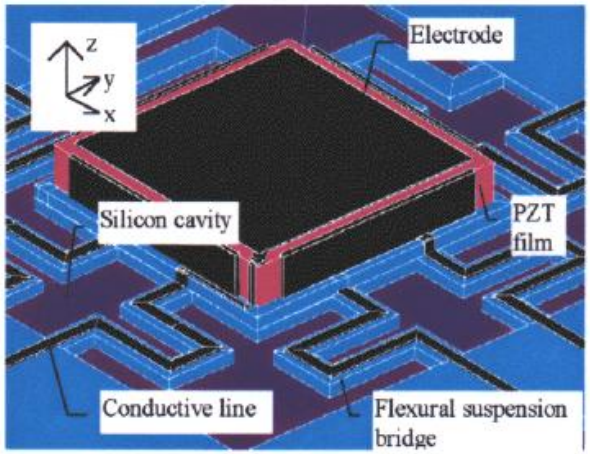
Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Digital Gyroscope

- Draper Tuning Fork Gyro
  - rotation of fork creates Coriolis Force
  - detected through either electrostatic, electromagnetic or piezoelectric.



- Piezoelectric Gyroscopes
  - Piezoelectric plate with vibrating thickness
  - Coriolis effect causes a voltage form the material



Clifton, D. (n.d.). Gyroscope [PowerPoint presentation]. Northwestern University. http://clifton.mech.northwestern.edu/~me381/project/done/Gyroscope.ppt

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Summary

Feature	Draper Tuning Fork Gyro	Piezoelectric Gyroscope
Temperature Sensitivity	No change in performance due to temperature	Affected by temperature variations
Voltage Noise	Lower voltage noise	Higher voltage noise in certain conditions
Signal to Noise Ratio	Stronger signal to noise ratio	Weaker signal to noise ratio
Communication with External Devices	Better communication capabilities	May require additional processing
Sensitivity	Higher sensitivity	Less sensitive
Input Voltage	Typically higher input voltage	Lower input voltage than vibrating mass
Measurement Capability	Measures rotation effectively	Measures rotation in two directions with a single device
Electronic Orientation Adjustment	Limited electronic adjustment options	Adjusting orientation electronically possible
Output Characteristics	Stable output under various conditions	Output is large when $\Omega = 0$

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Location Base Services

Most materials are from books and online sources. For use in lectures only. Not for duplication.

Class discussions:

What are the applications of LBS?

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

What's is Location Base Service

- A **location-based service** (LBS) is an information and entertainment service accessible via mobile devices that uses the device's geographical position.

- Pull - customer initiates mobile positioning when requesting a service.
- Push- service provider requests mobile positioning, not the customer
- Tracking - customer or service requests the location of a mobile terminal

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Applications of Location base Services

- Emergency services
- Mobile advertising
- Location sensitive billing
- Fraud protection
- Asset tracking
- Fleet management
- Intelligent transportation systems
- Mobile yellow pages
- parcel tracking and vehicle tracking services
- Location base coupons or advertising
- personalized weather services and
- location-based games
- Requesting the nearest business or service, such as an ATM or restaurant
- Locating people on a map displayed on the mobile phone
- Receiving alerts, such as notification of a sale on gas or warning of a traffic jam

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Applications of Location base Services

- Resource tracking with dynamic distribution.
  - Taxis, service people, rental equipment, doctors, fleet scheduling.
- Resource tracking.
  - Objects without privacy controls, using passive sensors or RF tags, such as packages and train boxcars.
- Finding someone or something.
  - Person by skill (doctor), business directory, navigation, weather, traffic, room schedules, stolen phone, emergency calls. Targeted advertising, buddy list, common profile matching (dating)

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Values of LBS

- Use as a filter
  - to restrict results of search that are close enough to the customers' location
- Use as a pointer
  - to show the location of the customers as a dot on a map in tracking services
- Use as a definer/launcher for "area alarm
  - i.e. to launch a pre-defined activity (e.g. send a MSMS) with customer enters/leaves a certain area



Courtesy: Dr. W

ures only. Not for duplication.

What are the issues/problems of LBS?

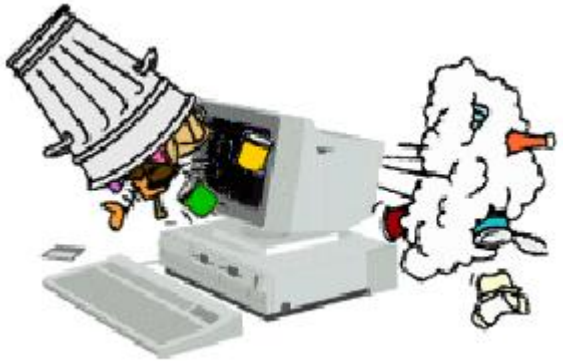
- Privacy
- Accuracy
- Coverage
- Useless information (Garbage information)
- Information overload

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Major issues faced by LBS

- Accuracy of location
  - Mobile network was primarily built for communication, not positioning

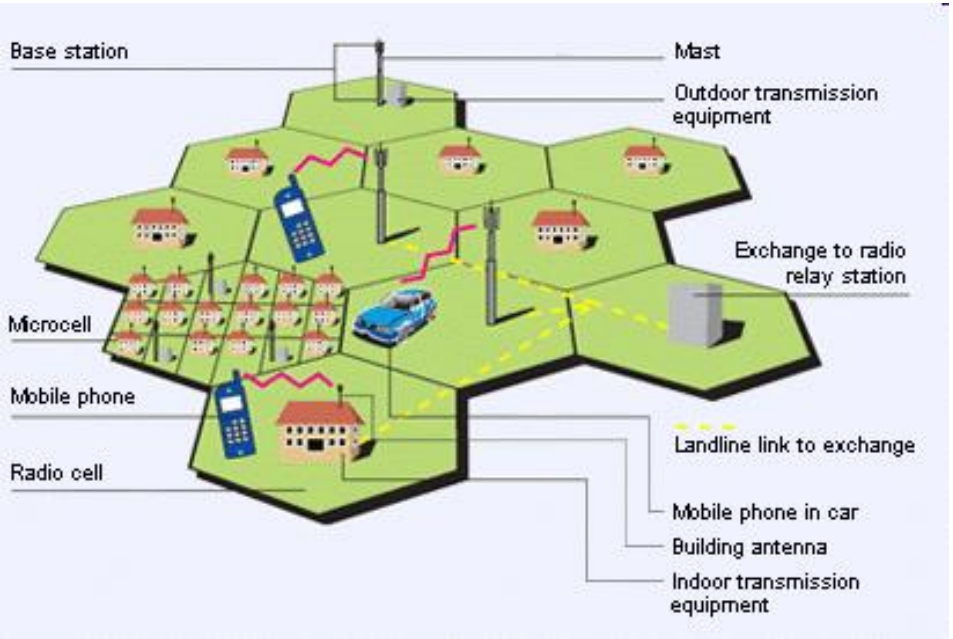


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Major issues faced by LBS

- Investment vs penetration
  - Extraction of network measurement for position calculation remains a difficult and expensive task
  - **More network penetration -> more expensive**



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Major issues faced by LBS

- Business Model (Willingness to pay)
  - Consumer market: **not essential. They can use GPS**
  - Enterprise users are more willing to pay a higher fee

- Inter-operability between operators
  - Cross border continuity
  - Difference in positioning accuracy
  - Commercial models in roaming

Privacy

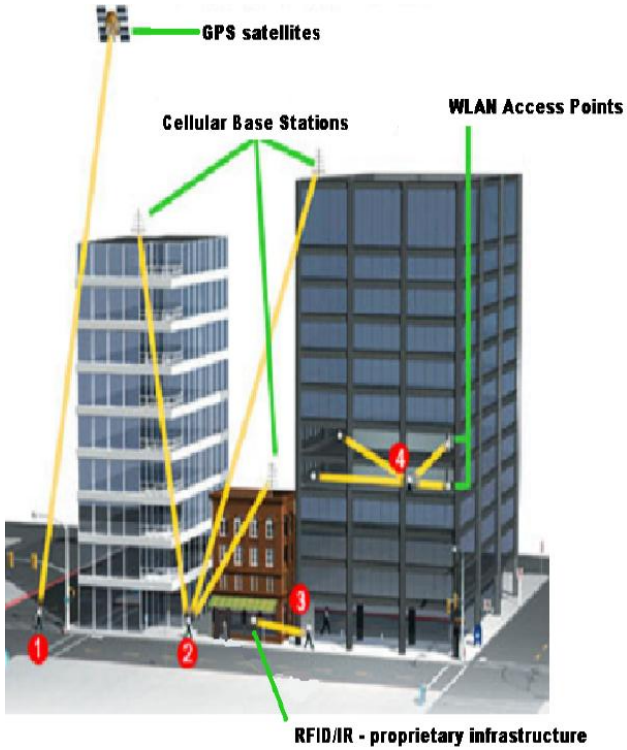
Opt-in/ Opt-out policy

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Enabling technologies

- Location base technologies
  - Global Positioning System GPS
  - RFID Radio Frequency ID
  - Mobile Phone Location Base
  - WiFi Location Base

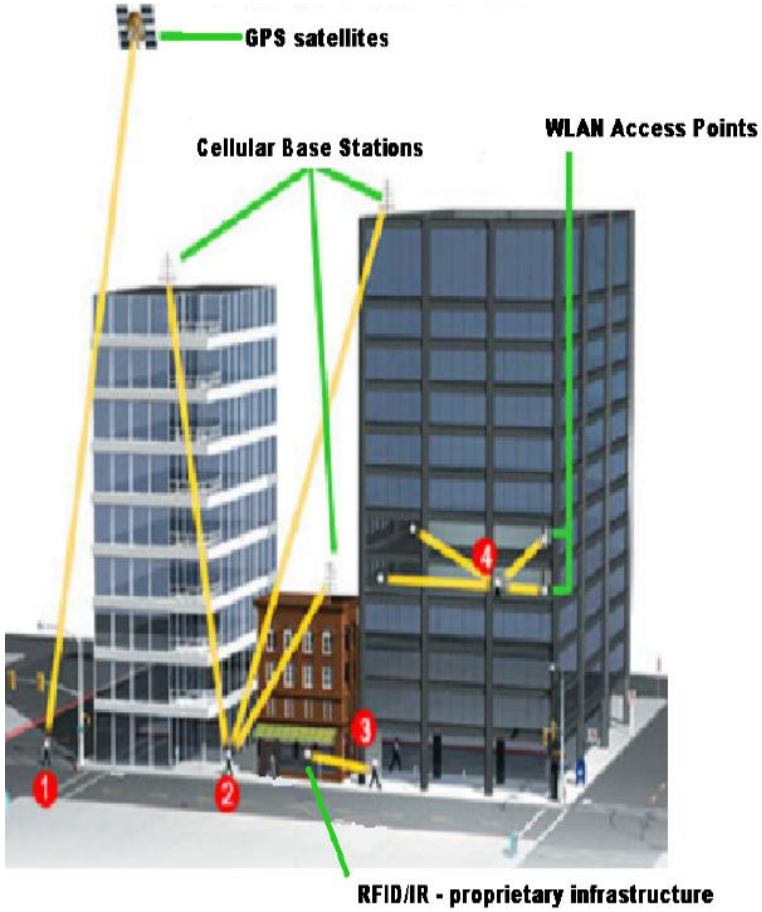


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Location base technologies

- GPS
  - Accuracy within 2-20m
  - Does not work well indoors and in the urban canyons of large cities
- RFID
  - Needs special readers, tags
  - Can be used to locate object passing close to RF reader



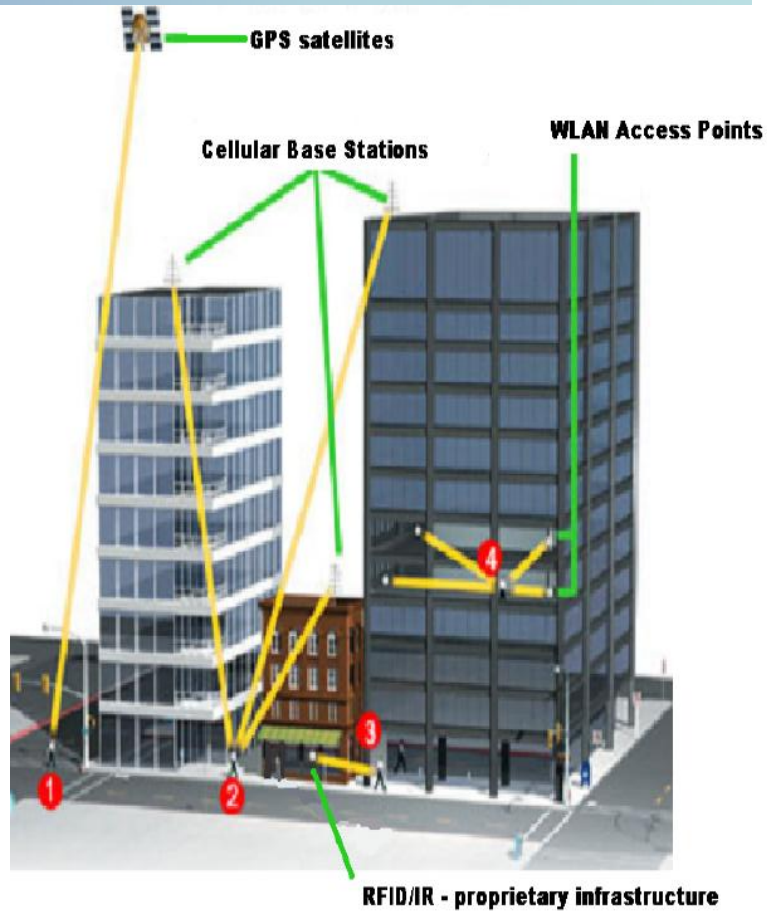
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Location base technologies

- Mobile phone
  - At best 100m accuracy
  - Not adequate accuracy for indoor tracking

- WLAN/WiFi
  - Cell-ID: Poor accuracy, Cannot differentiate one floor from the other
  - WLAN proprietary “location” access points: Needs special infrastructure and Costly, difficult to deploy



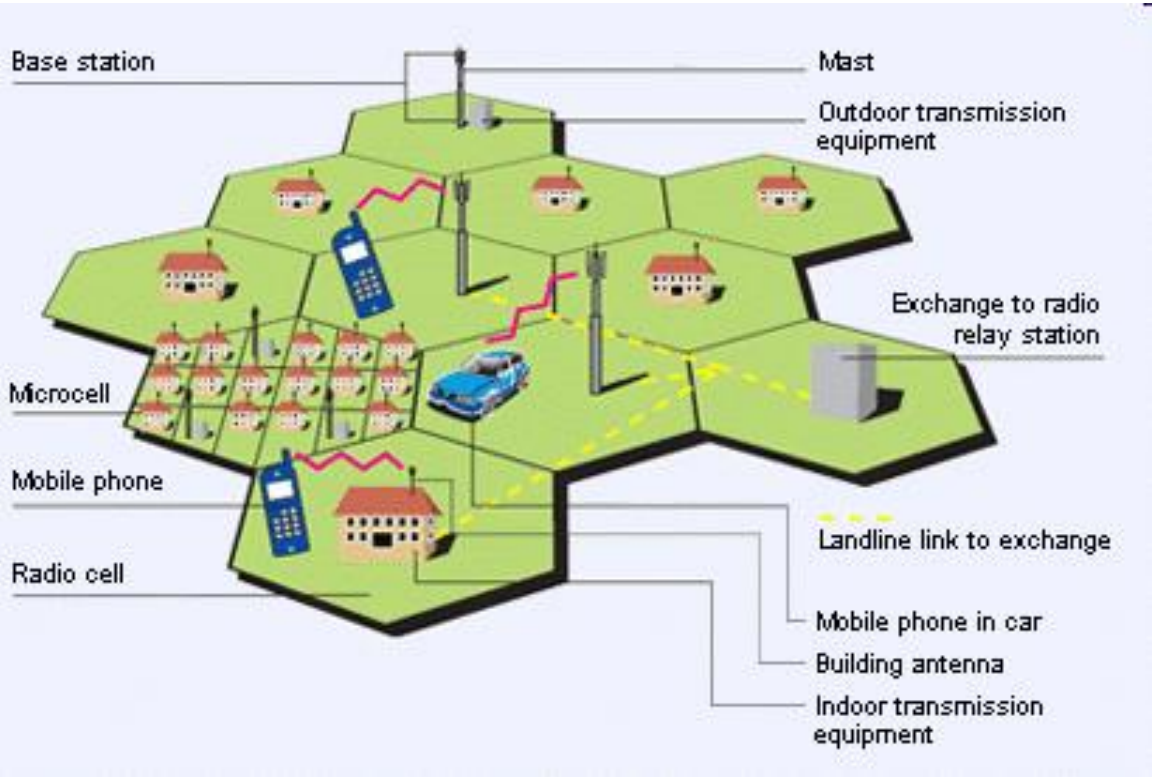
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Enabling Technology: Location Base Service

- Cellular radio network is the foundation for location base services

- GSM is a cellular network
  - → mobile phones connect to it by searching for cells in the immediate vicinity.



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Mobile Communication

31

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Celluar Network



A



B



C



D

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

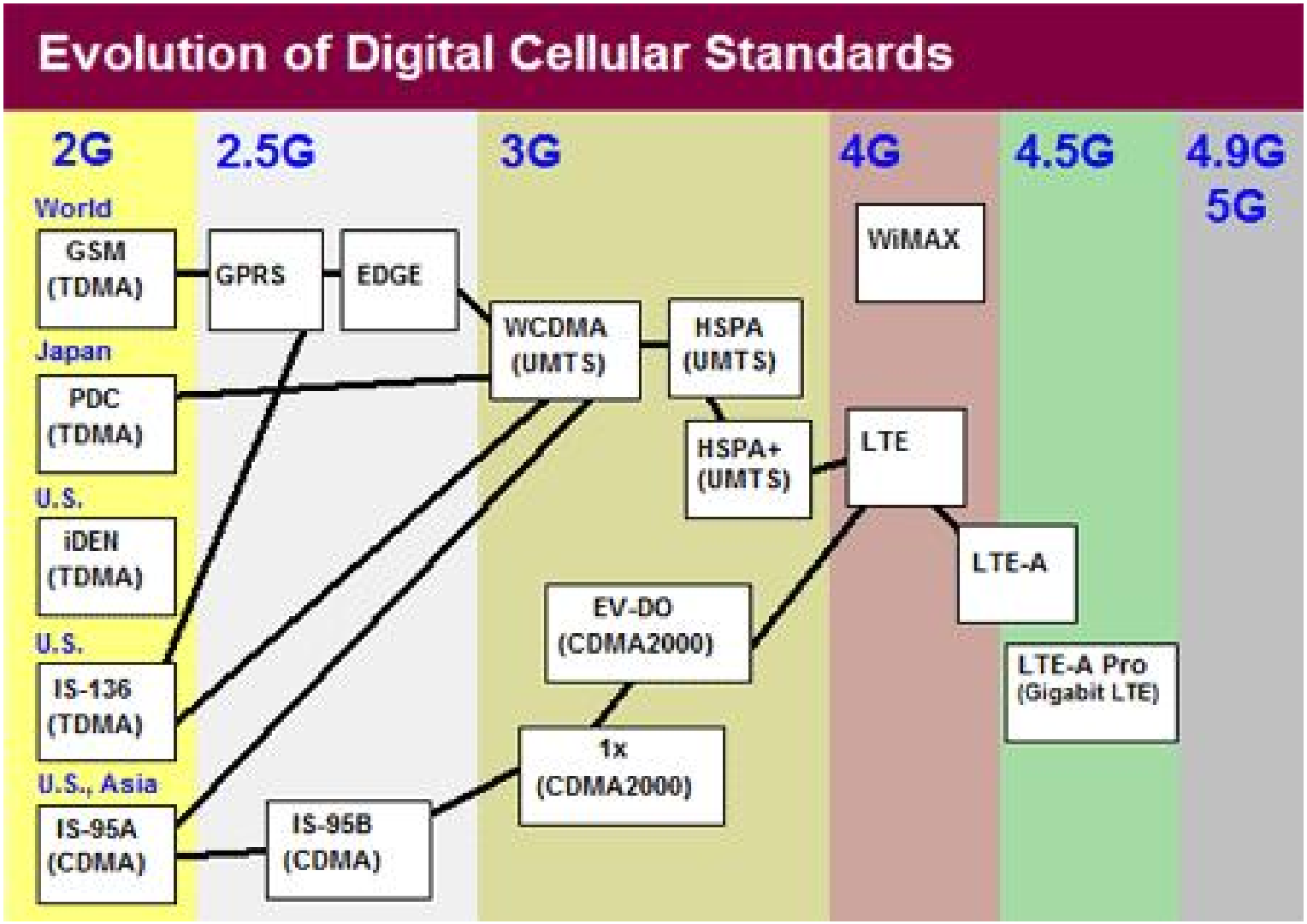
Celluar Network

- **0G - Briefcase-size mobile radio telephones**
  - no cellular structure
- **1G - Analog cellular telephony**
  - focusing on voice calls.
- **2G- Digital cellular telephony**
  - enabling SMS and basic data services.
- **3G- High-speed digital cellular telephony**
  - supporting video calls and mobile internet.
- **4G - IP-based mobile network**
  - enabling high-definition multimedia streaming and robust internet access.
- **5G- Fifth-generation mobile networks**
  - Extremely high speeds, low latency, and support for massive IoT connectivity.
- **6G - Sixth-generation mobile networks** (in development).
  - Aims for terahertz frequencies, AI integration, and advanced applications like holographic communication.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Celluar Network



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Celluar Network

Feature	2G (CDMA/TDMA)	3G (CDMA2000/WCDMA)	4G/5G (LTE)
Technology Type	Circuit-switched, TDMA/CDMA	Packet-switched, CDMA2000/WCDMA	All-IP, OFDM
Data Rates	Up to 64 kbps	144 kbps (mobile), up to 2 Mbps (fixed)	100 Mbps (4G), 1 Gbps (5G)
Transmission Method	Time Division (TDMA) / Code Division (CDMA)	Code Division (CDMA)	Orthogonal Frequency Division Multiplexing (OFDM)
Capacity	Limited capacity due to circuit switching	Improved capacity with packet switching	High capacity with efficient spectrum use
Latency	Higher latency	Moderate latency	Low latency (sub-10 ms)
Coverage	Good	Improved	Best
Mobility	Limited	Enhanced	Seamless
Applications	Voice-centric services	Voice and data services	High-speed internet, video streaming, IoT
Deployment	Primarily in urban areas	Broader deployment, including rural	Global, with extensive deployment across various bands

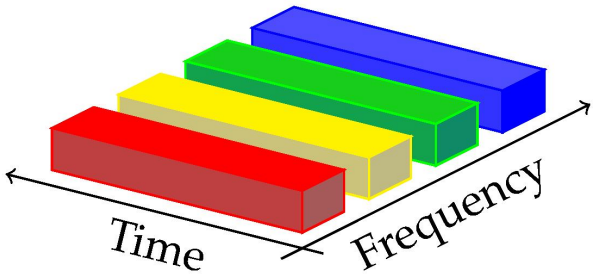
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Celluar network

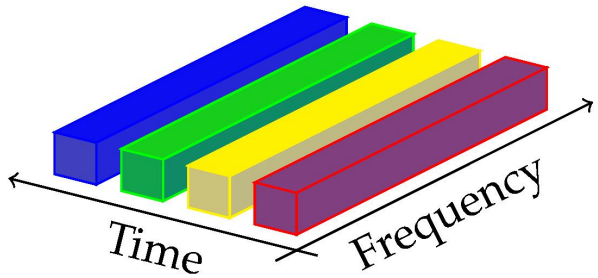
1G

FDMA



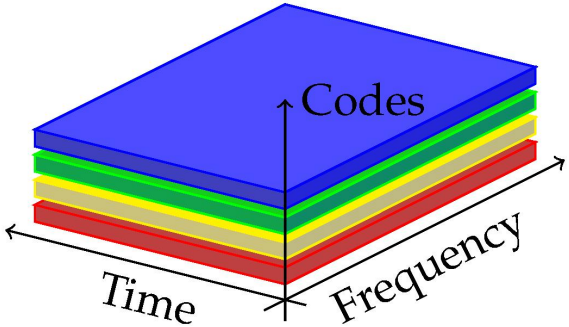
2G

TDMA



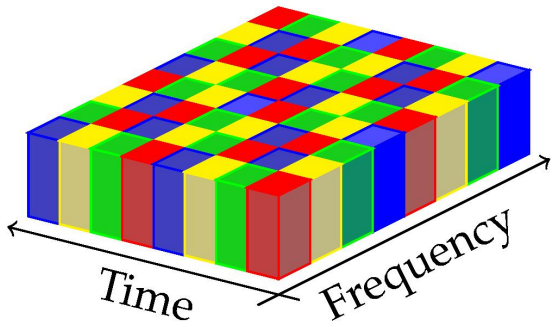
3G

CDMA



4G

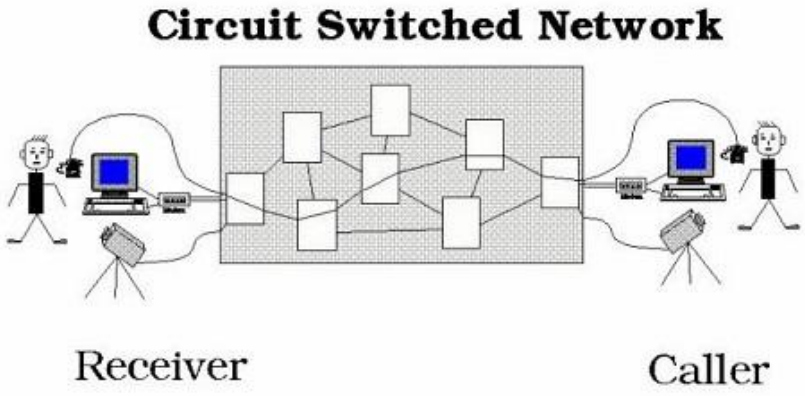
OFDM

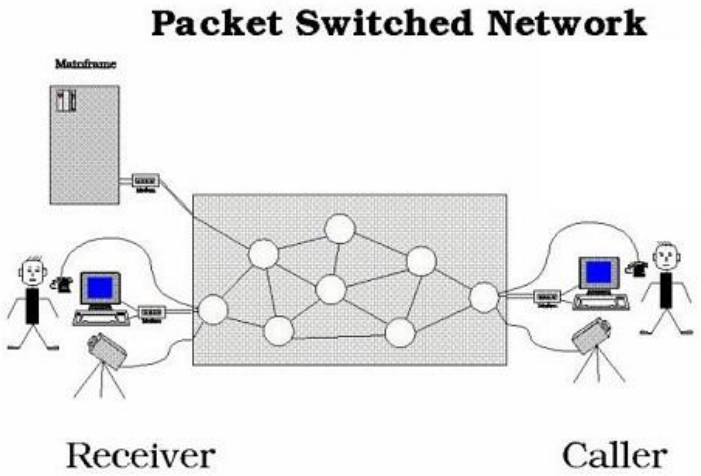


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Celluar network





Circuit switching	Packet Switching
A physical path is established between the two end points.	Data is divided into small units called packets with each packet carrying reciever/caller location
Less efficient	More efficient, allowing multiple connections at the same time

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GSM

- **GSM (Global System for Mobile Communications** is/was the most popular standard for mobile telephony systems in the world.
- GSM is considered a **2G mobile phone system**.
- It facilitates the wide-spread implementation of data communication applications into the system.
- Advantages
  - Roaming
  - Switch carriers without replacing phones
  - Support SMS
  - Support SIM Card
  - Support a worldwide emergency telephone number feature 112

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

SIM Card in GSM

- **SIM Card = Subscriber Identity Module**
- A detachable smart card containing the user's subscription information and phone book.
- This allows the user to retain his or her information after switching handsets/changing operators
- Some mobile phone operators **restrict handsets** that they sell for use with their own network.
- Because operators must **recoup the subsidized purchase price** of mobile phones from subscriptions





Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Network structure

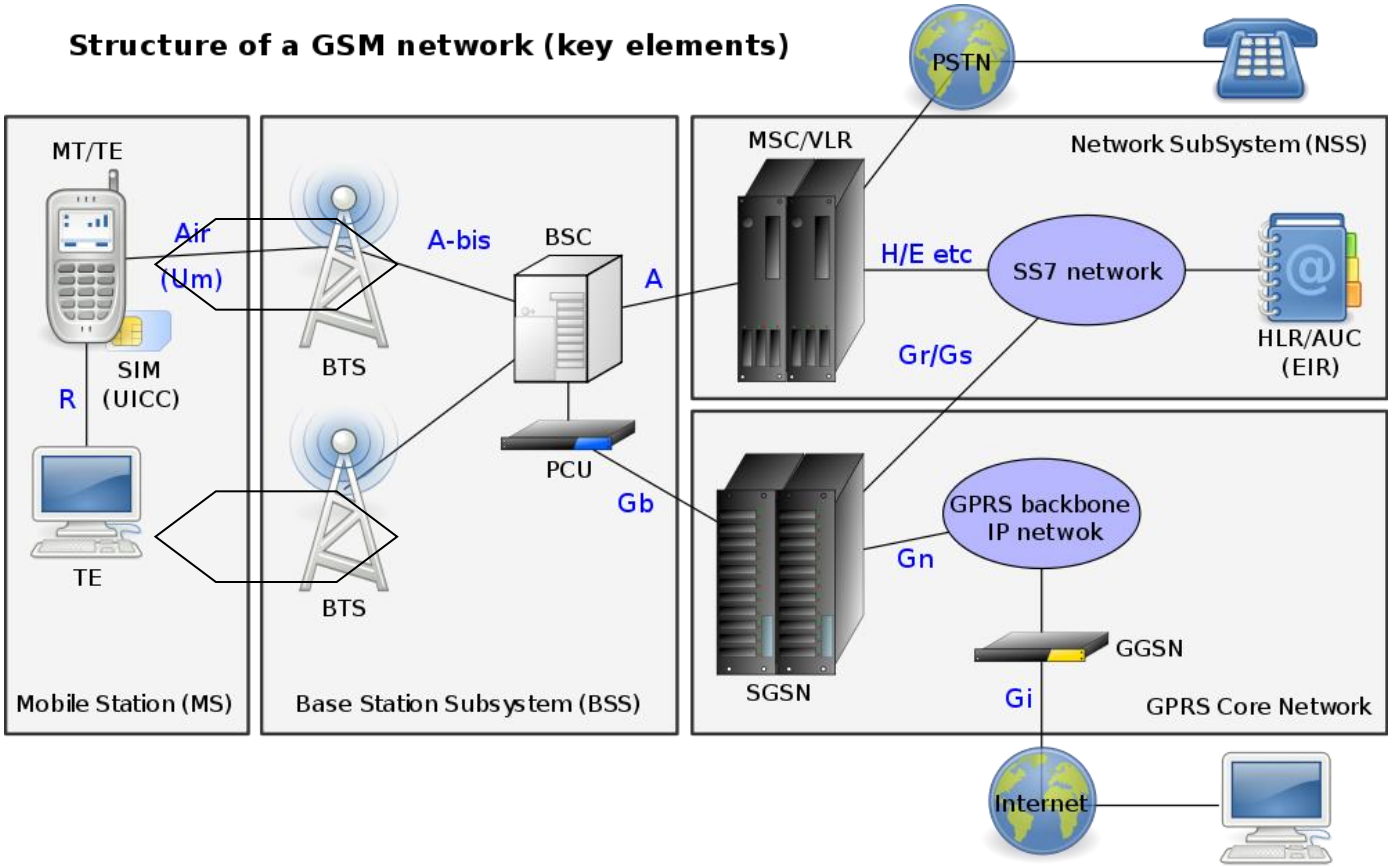
- The structure of a GSM network:
  - Base Station Subsystem
    - base stations and their controllers
  - Network and Switching Subsystem
    - part of the network most similar to a fixed network). This is sometimes also just called the core network.
  - GPRS Core Network
    - the optional part which allows packet based Internet connections
  - Operations support system (OSS)
    - for maintenance of the network.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Network structure

- Base Station Subsystem
- Network and Switching Subsystem
- GPRS Core Network
- Operations support system (OSS)



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Network structure

- Base Station Subsystem
  - Manages the radio communication between the mobile devices and the network.
- Network and Switching Subsystem
  - Responsible for call routing, mobility management, and subscriber data management.
- GPRS Core Network
  - enabling mobile internet and data transmission.
- Operations support system (OSS)
  - Provides tools and systems for network management and maintenance.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Cell sizes and coverage

- There are 5 different cell sizes in a GSM network
  - Macro,
  - Micro,
  - Pico,
  - Femto and
  - Umbrella cells.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Cell sizes and coverage

- The coverage area of each cell varies according to the implementation environment.
  - Macro cells
    - Cells where the base station antenna is installed on a mast or a building above average roof top level.
  - Micro cells
    - Those antenna height is under average roof top level; they are typically used in urban areas.







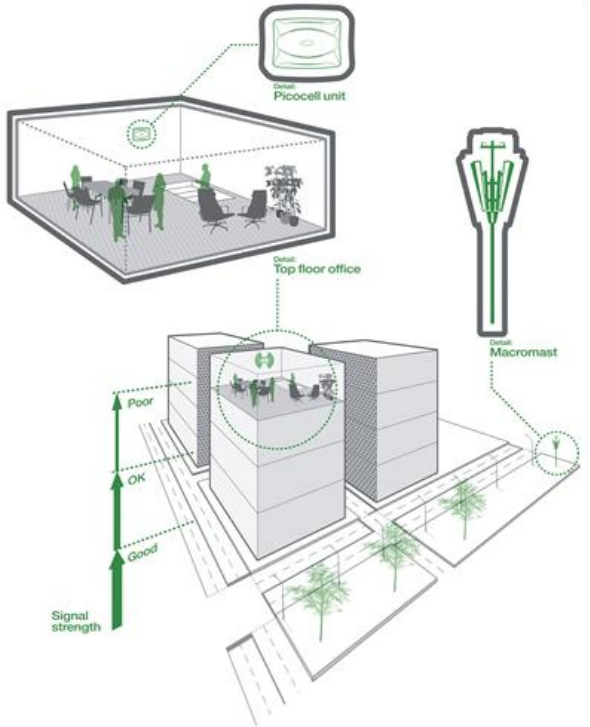


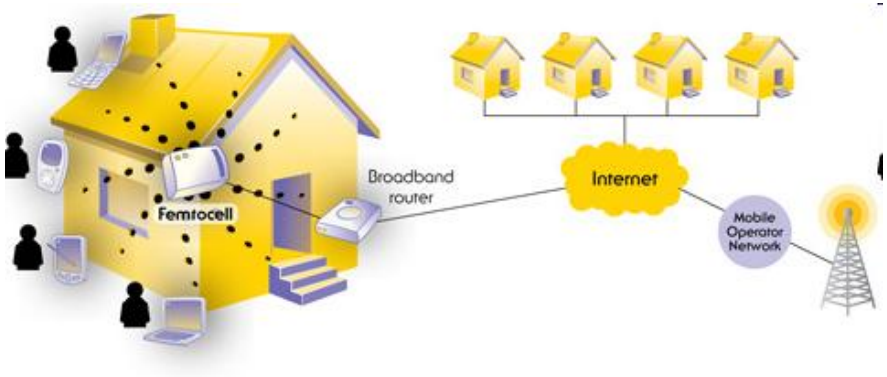
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Cell sizes and coverage

- Picocells
  - Small cells whose coverage diameter is a few dozen metres; they are mainly used indoors.
- Femtocells
  - Cells designed for use in residential or small business environments and connect to the service provider's network via a broadband internet connection.
- Umbrella cells
  - Used to cover shadowed regions of smaller cells and fill in gaps in coverage between those cells.





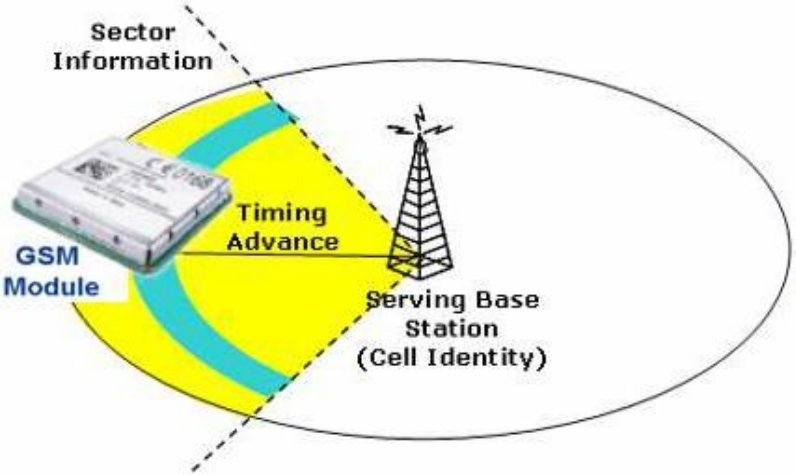


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Cell sizes and coverage

- Cell horizontal radius varies depending on:
  - antenna height,
  - antenna gain
  - propagation conditions from a 300m -35km



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Cell sizes and coverage

- Indoor coverage is also supported by
  - **indoor picocell base station**, or
  - an **indoor repeater** with distributed indoor antennas fed through power splitters,
  - to deliver the radio signals from an antenna outdoors to the separate indoor distributed antenna system.
  - E.g. In shopping centers or airports.



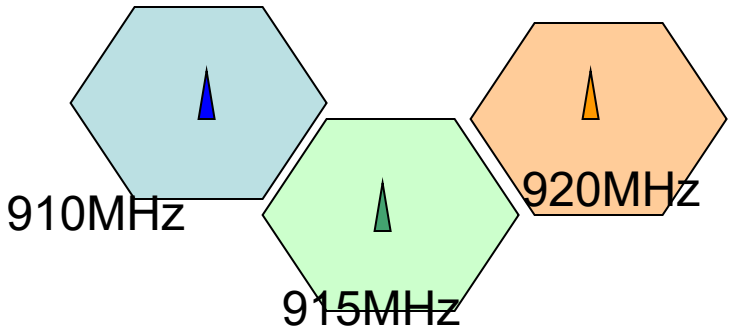


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Cellular Network

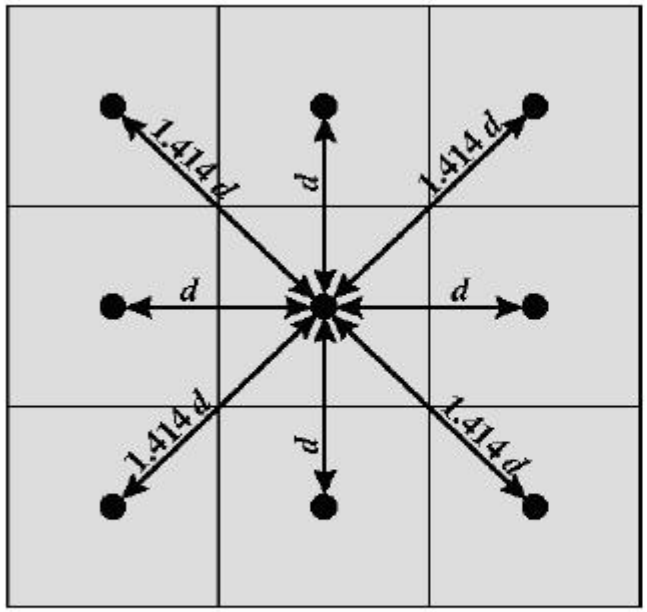
- Use multiple low-power transmitters (100 W or less)
- Areas divided into **cells**
  - Each served by its own antenna
  - Served by base station consisting of transmitter, receiver, and control unit
  - Band of frequencies allocated
  - Cells set up such that antennas of all neighbors are equidistant (hexagonal pattern)



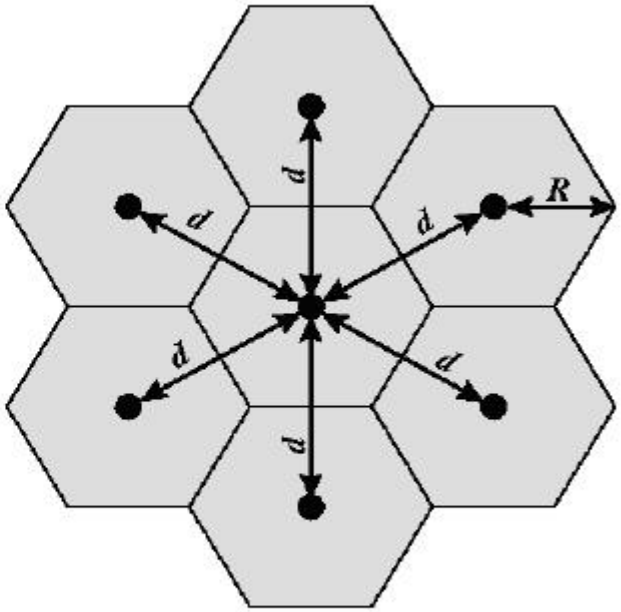
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Cell Format



(a) Square pattern



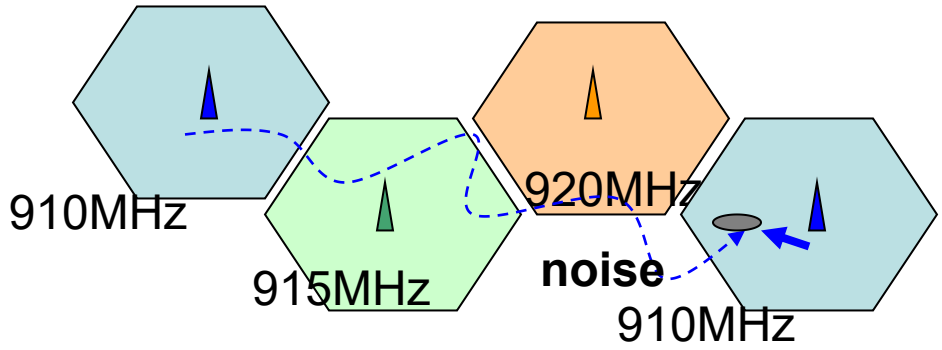
(b) Hexagonal pattern

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Frequency Reuse

- Adjacent cells assigned different
  - frequencies to avoid interference or crosstalk
- Objective is to **reuse frequency in nearby cells**
  - 10 to 50 frequencies assigned to each cell
  - Transmission power controlled to limit power at that frequency escaping to adjacent cells
  - The issue is to **determine the number of cells avoiding crosstalk**

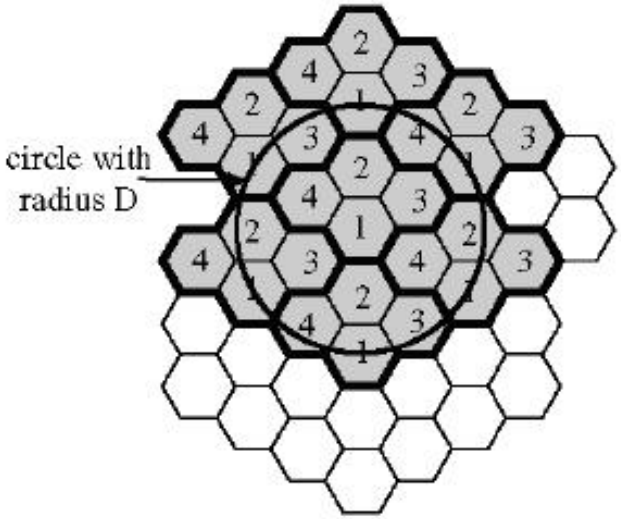


Courtesy: Dr. Wilton Fok

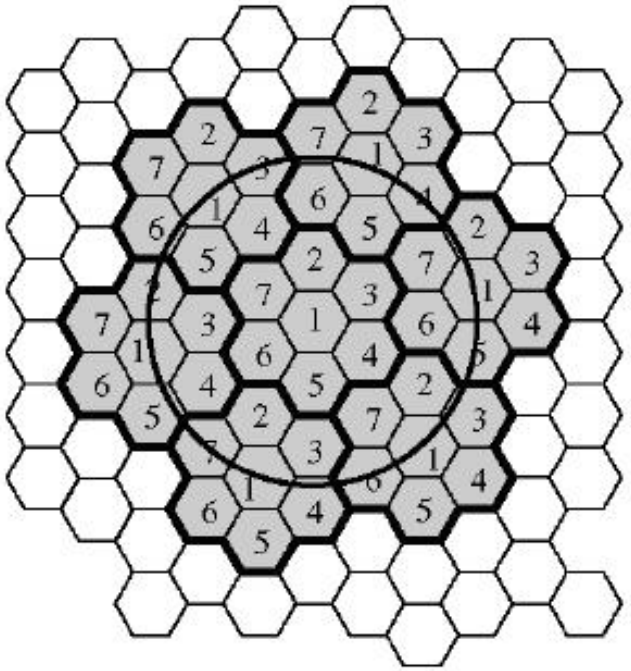
Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Frequency Reuse

- N=1,3,4,7,9,12,13,16,19 .. (Reuse factor/Cluster size)



(a) Frequency reuse pattern for $N = 4$



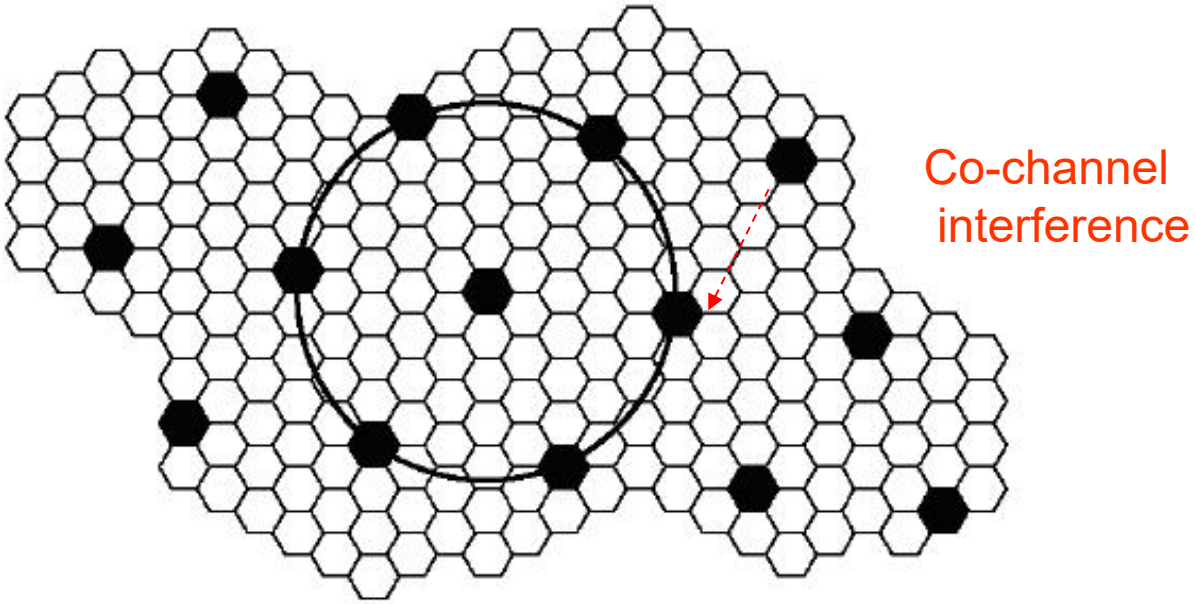
(b) Frequency reuse pattern for $N = 7$

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Frequency Reuse

- Larger N
  - Longer distance between cell with the same frequency
  - **Less co-channel interference**



(c) Black cells indicate a frequency reuse for $N = 19$

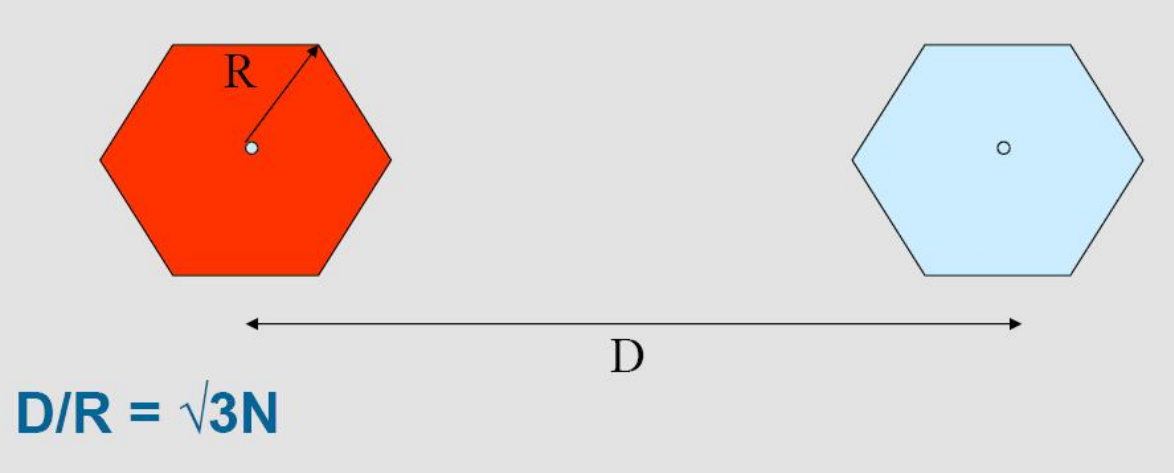
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Frequency Reuse

Power is **inversely proportional to Distance** (Inverse square law)

→ Signal to Noise is proportional to N^2



D: Distance between cell towers
R: Radius of of a cell
N: Reuse factor/Cluster size

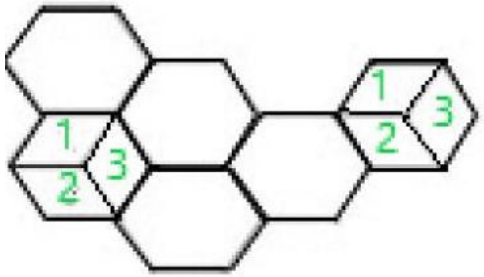
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

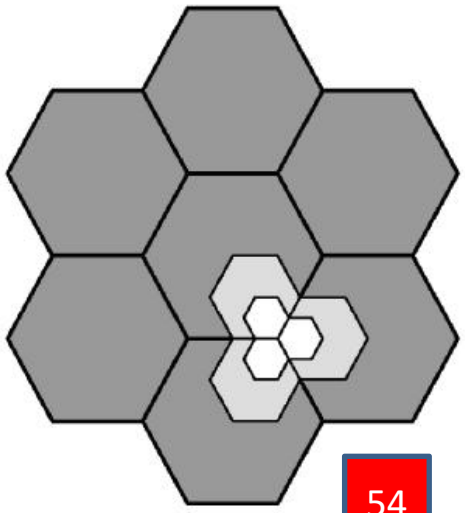
Approaches to Cope with Increasing Capacity

- Adding new channels
- Frequency borrowing
  - frequencies are taken from adjacent cells by congested cells
- Cell splitting
  - cells in areas of high usage can be split into smaller cells
- Cell sectoring
  - cells are divided into a number of wedge-shaped sectors, each with their own set of channels
- Microcells
  - antennas move to buildings, hills, and lamp posts

Cell Sectoring



Cell Splitting

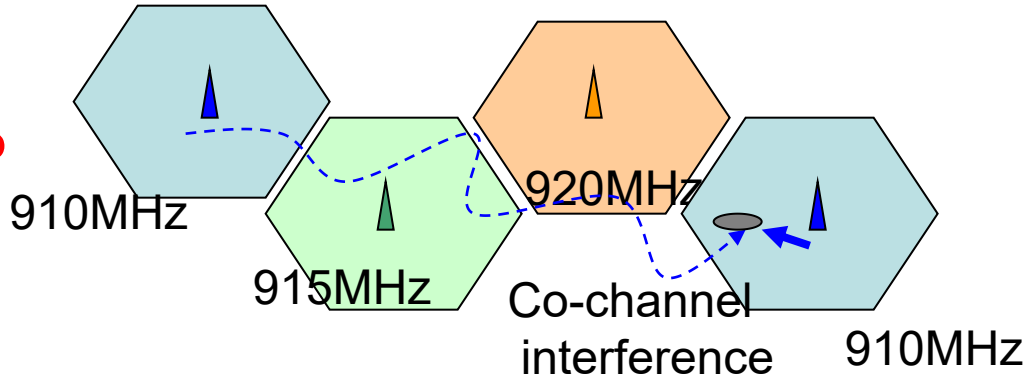


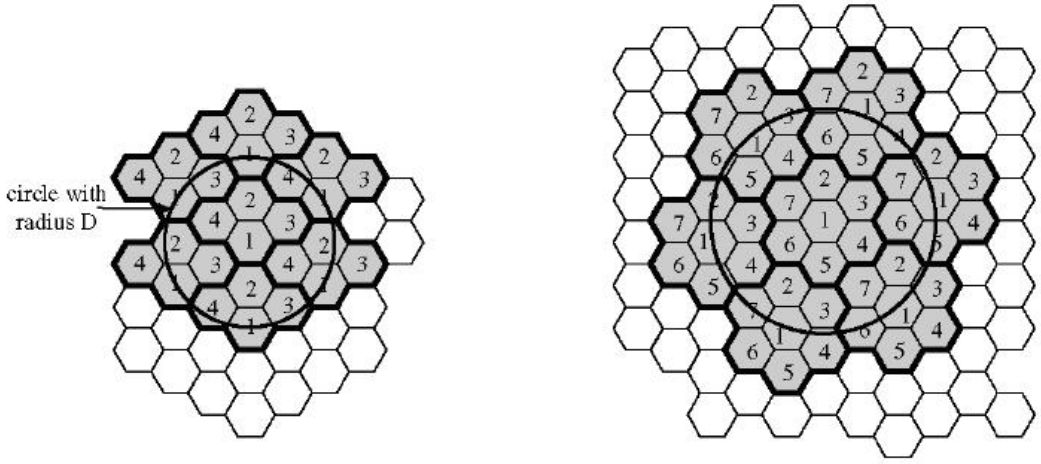
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Capacity vs Accuracy

- When N is large
  - **Less co-channel interference**
  - **Stronger signal to noise ratio**
  - Smaller cell-site
  - More accurate in location estimation
- But require more frequency resources





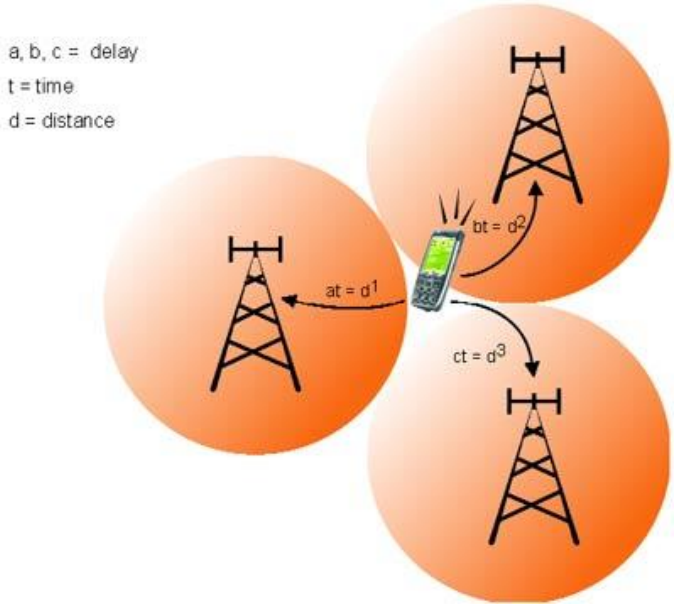
(a) Frequency reuse pattern for $N = 4$ (b) Frequency reuse pattern for $N = 7$

Courtesy: Dr. Wilton Fok

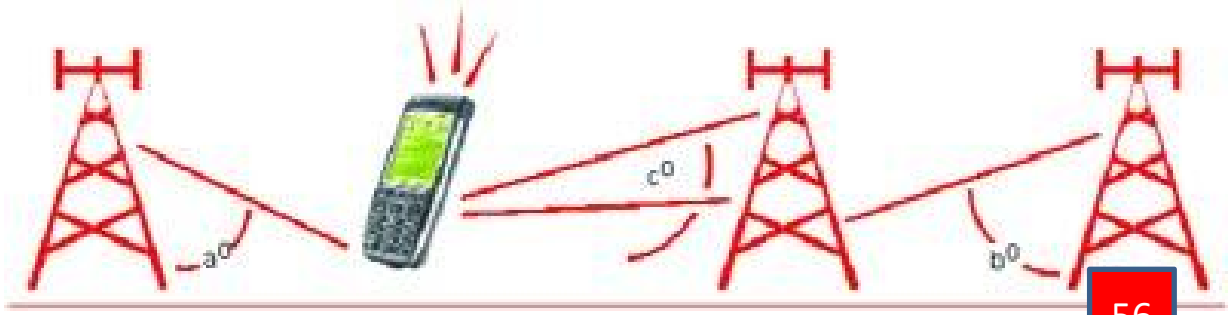
Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Methodologies for location estimation

- Time of arrival (TOA)
  - Time predict distance
  - 3 distance data locate a point



- Angle of arrival (AOA)
  - 3 angles location a point

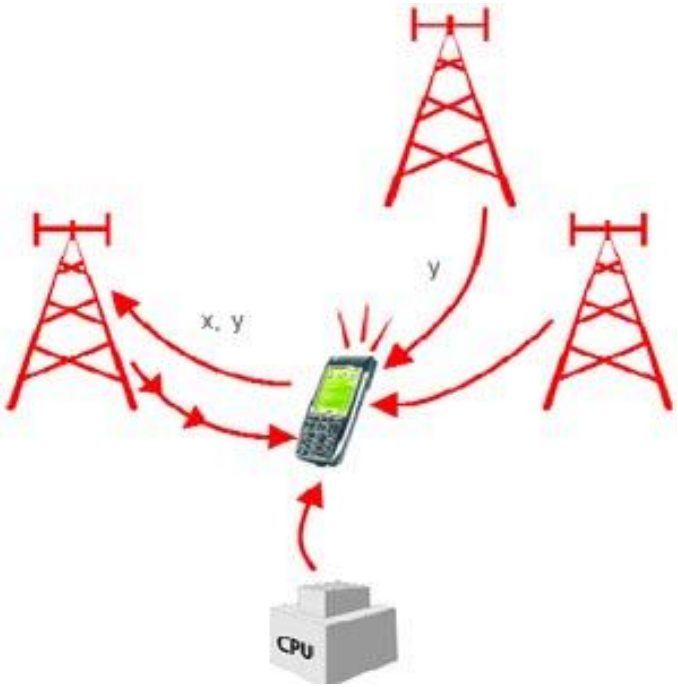


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Methodologies for location estimation

- **Time difference of arrival (TDOA)**
  - If it can only access two Base-station, distance between base-stations provide the 3rd information

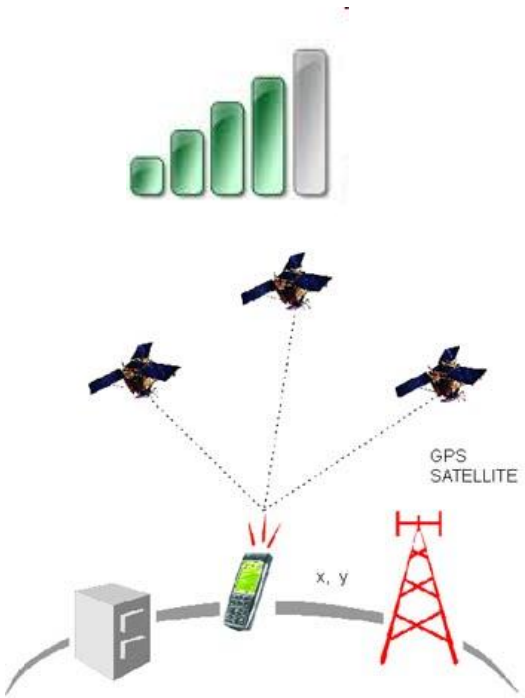


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Methodologies for location estimation

- **Enhanced observed time difference (EOTD)**
- **Cell global identification (CGI) and Timing advance (TA)**
- Signal strength (SS)
- Global Positioning System (GPS)
- Control Plane Locating



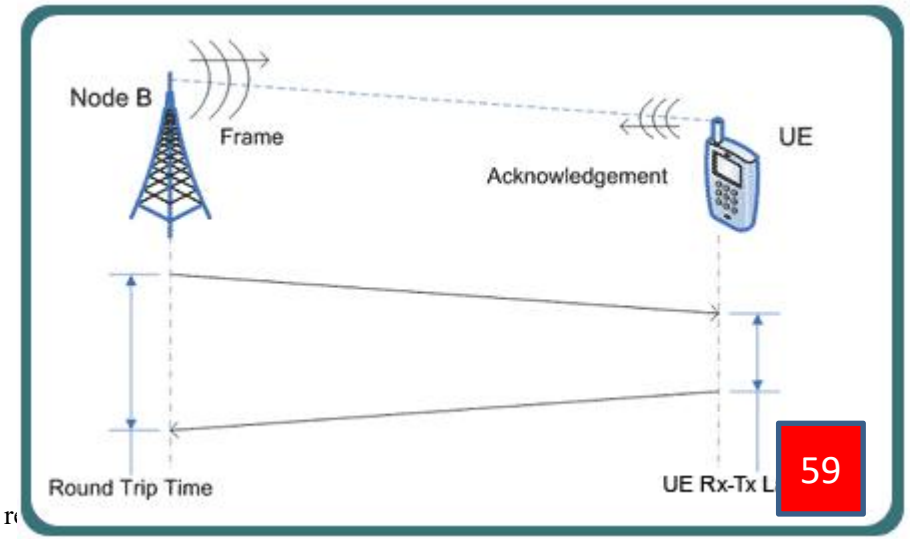
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Methodologies for location estimation

- Control plane locating
  - Service provider gets the location based on the **radio signal delay** of the closest cell-phone towers (for phones without GPS features)
  - **which can be quite slow** as it uses the 'voice control' channel.
  - Some primitive LBS services use a single base station, with a **'radius' of inaccuracy**, to determine a phone's location.

- More details
  - http://www.hkwtia.org/



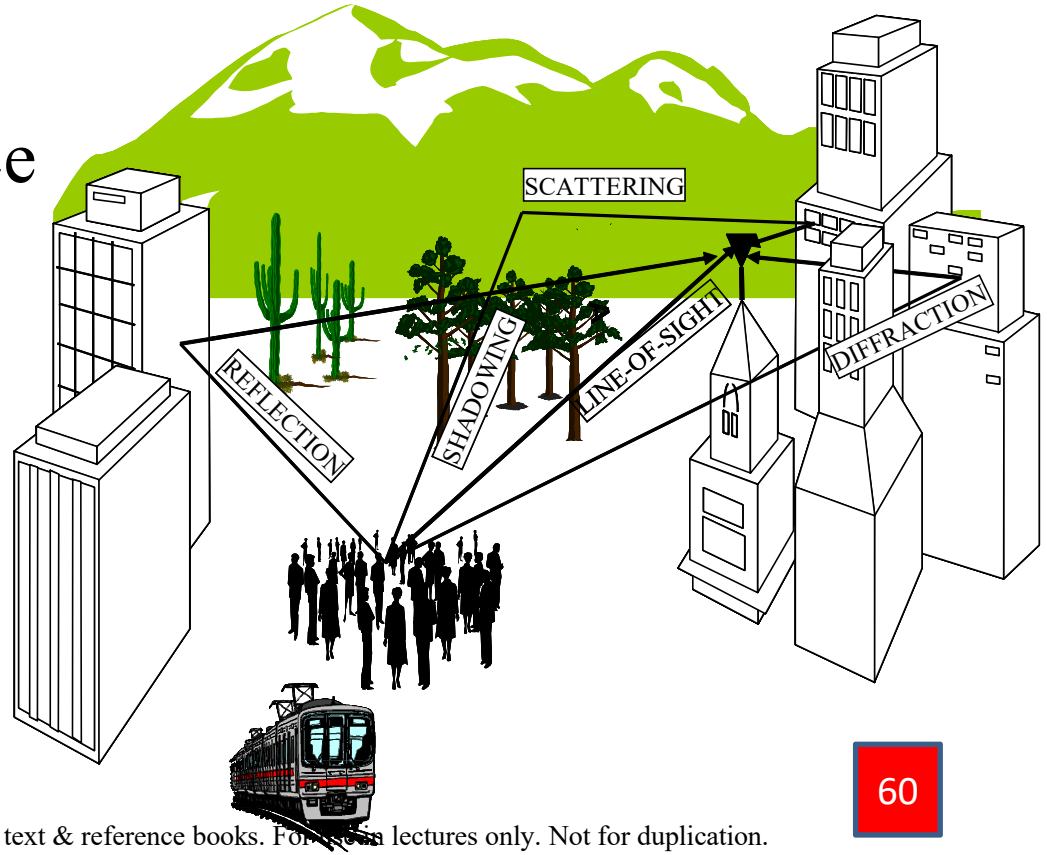
Courtesy: Dr. Wilton Fok

Most materials are from the text & r

Error of location base estimation

- Scattering
- Shadowing
- Reflection
- Diffraction
- Co-channel interference





Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For [illegible] lectures only. Not for duplication.

Global Positioning System
GPS

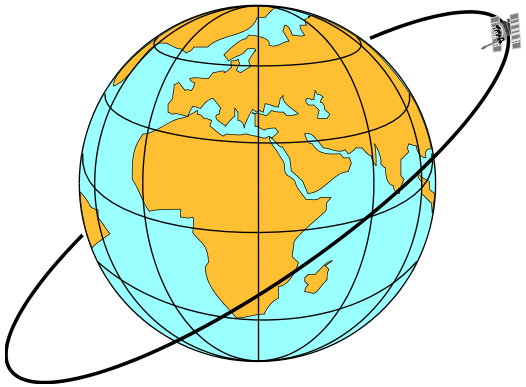
61

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Background

- GPS is a satellite based navigation system offering precision navigation capability.
- Originally designed for military use, civilian access has been permitted to specific parts of the GPS.



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Background

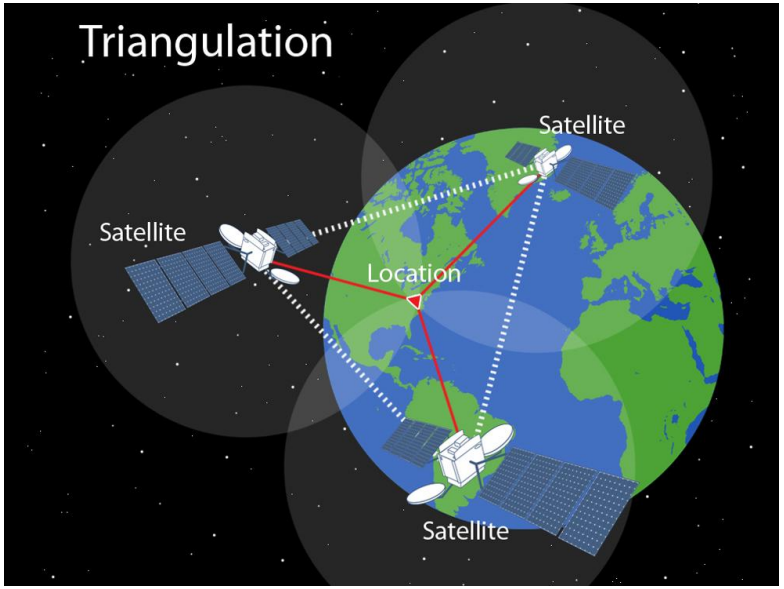
- Features
  - **Position accuracy of 10-100 m** or better in three dimensions.
  - **available 24 hours per day** throughout the world and in all weather conditions.
  - **resistance to** intentional (jamming) and unintentional **interference**.
  - GPS signals is affordable and reliable and does not require atomic clocks or antenna arrays.
  - For the GPS user, the system is passive and **requires a receiver only** without the requirement to transmit.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Operation principle

- By triangulation
  - By knowing the **position of the satellite** and the **distance from the satellite**
  - **Combinations of satellites** can be used to determine the exact position of the receiver.



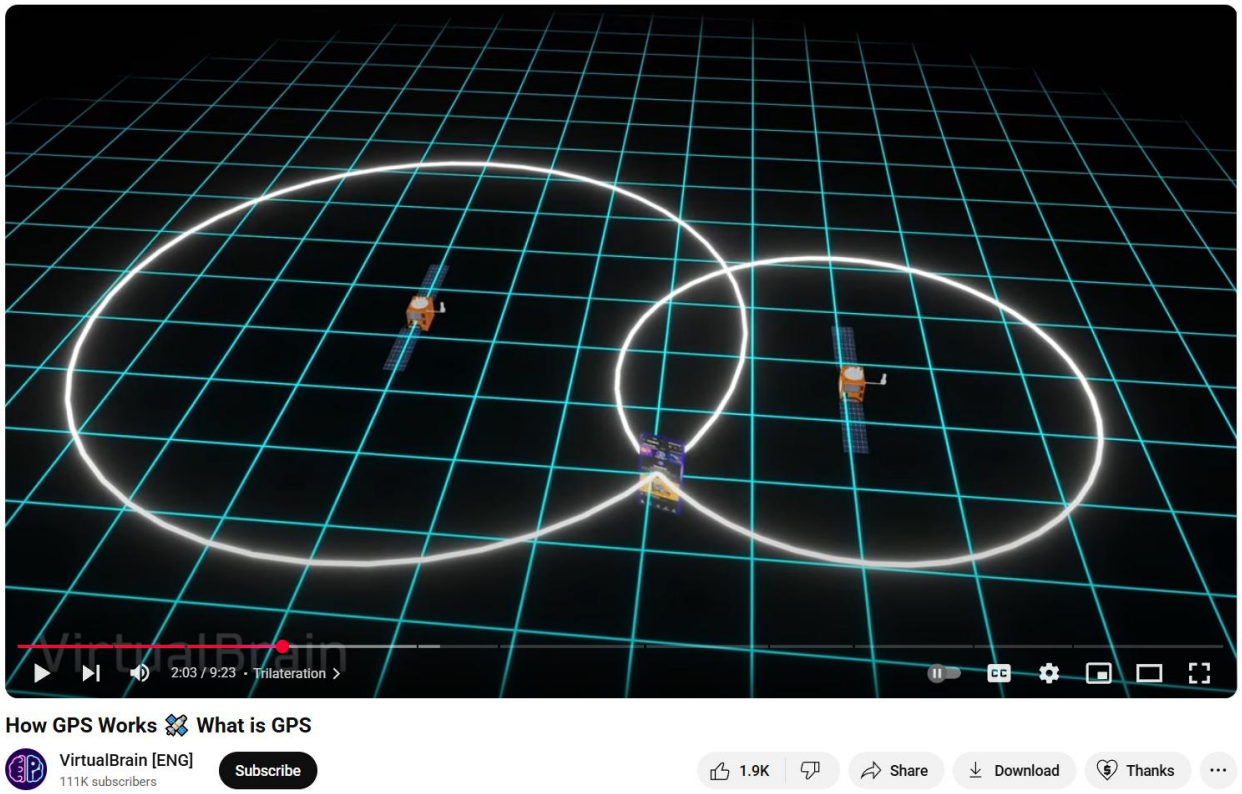
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Operation principle

How GPS works

https://youtu.be/AlHPDRQ08jU?t=104

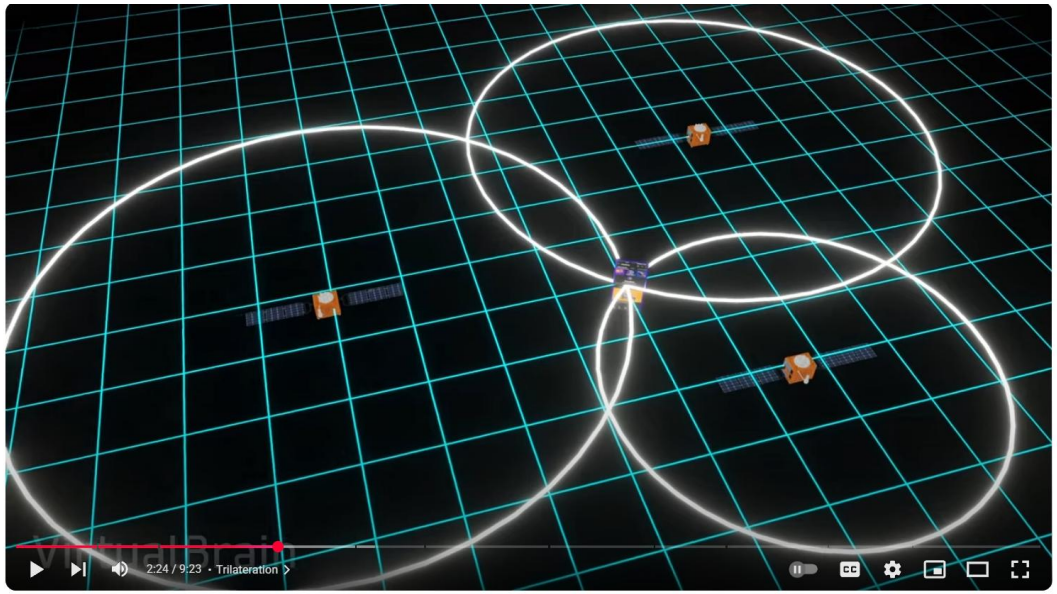


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Operation principle

- With 1 satellite, and knowing the position of that satellite, the location of the user would be anywhere along an arc.
- If 3 satellites were used, the location of the user would be at the intersection of the 3 arcs created by the satellites, as shown.



Satellite	Reception Time	Time Difference (sec)	Distance (km)
1	100.0855	0.0855	25650
2	100.0900	0.0900	27000
3	100.0875	0.0875	26250

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Operation principle

- Mathematical principle
  - To solve for the 3 dimensional position (with 3 variables: **latitude, longitude and altitude**), 3 equations (or satellites) are needed.

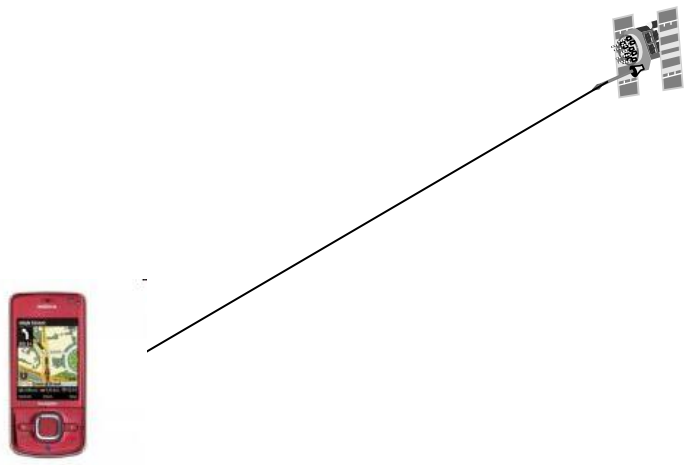
$$f1(x,y,z) = C1$$
$$f2(x,y,z) = C2$$
$$f3(x,y,z) = C3$$

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Operation principle

- A simplified GPS system illustrates the concept of satellite ranging.
- **A satellite transmits a time signal**, as shown. The **receiver is stationary.**
- By measuring the **time difference** from when the **signal left the satellite** to when it is **received by the user**, the distance can be calculated.
- Distance = time difference and the speed of light (300,000 km/sec).



distance = time x speed of light
i.e if time difference is 0.09 sec
distance = 0.09 sec x 300,000 km/sec
distance = 27,000 km

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

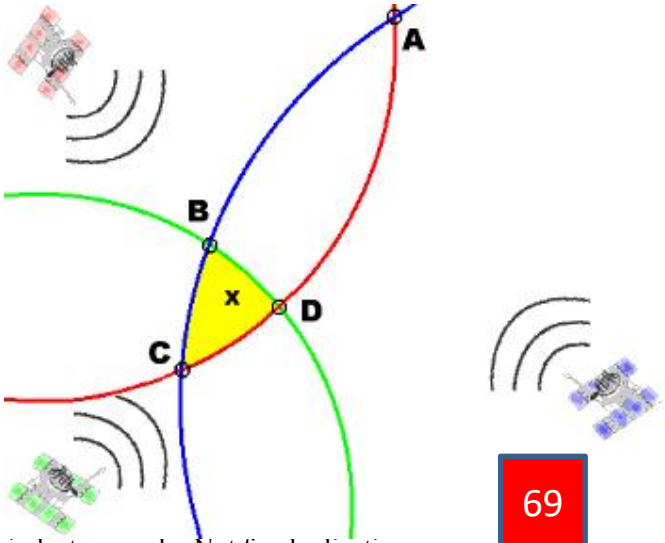
Problem - Clock bias

- **It assumes receiver clock in perfect synchronization** with the satellite and exhibiting the same accuracy.

- → Receiver clocks are not perfectly synchronized satellite time.
  - For every microsecond (one-millionth of second) difference between the satellite clocks and the receiver clock, a 300 meter error is introduced.

- This error is known as a clock bias, **an extra satellite is required**

$$f1(x,y,z,\varepsilon) = C1$$
$$f2(x,y,z,\varepsilon)) = C2$$
$$f3(x,y,z,\varepsilon)) = C3$$
$$f4(x,y,z,\varepsilon)) = C4$$



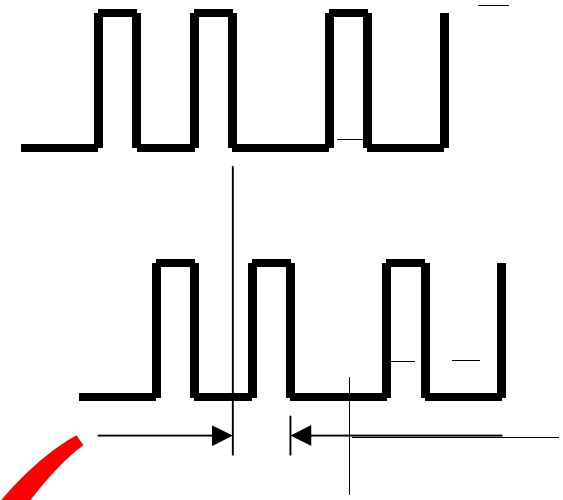
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

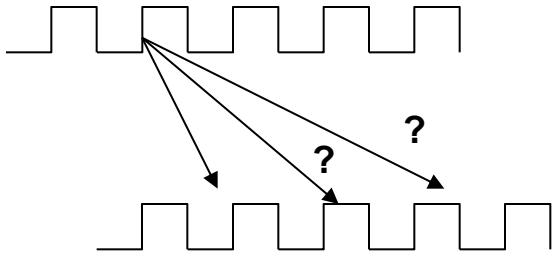
GPS technical: modulation

- The GPS signal modulation consists of a repetitive binary signal with special pattern to determine the time.
- The **special pattern allows the time difference to be identified**.
- The waveform from the satellite is matched that of the receiver to **compute the distance from a satellite**.

Repetitive binary signal with special pattern



Simple periodic binary pulse

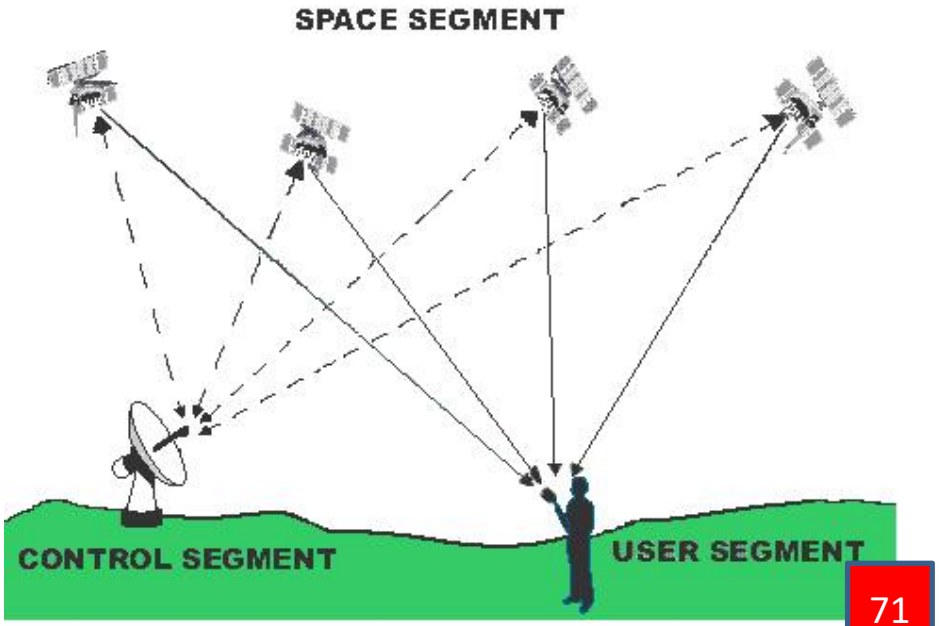


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Components

- GPS consists of 3 major components:
  - GPS Satellites,
  - Control segment,
  - User (GPS Receiver)



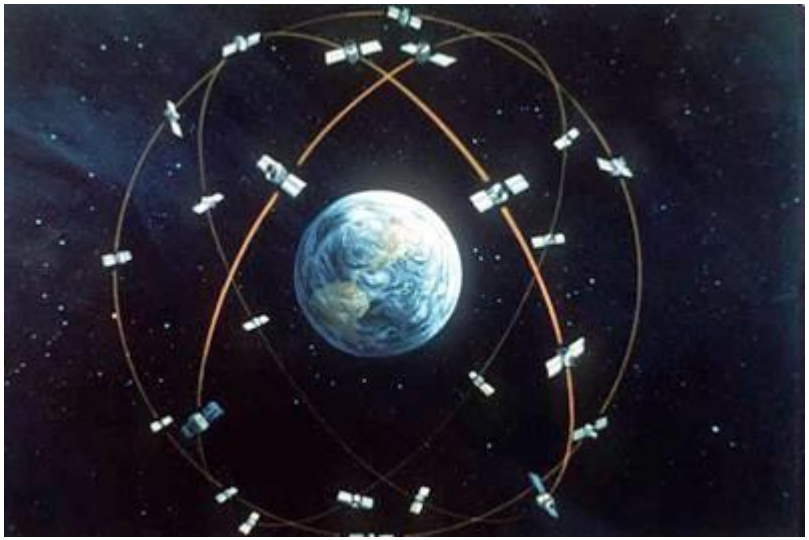
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Satellites

- There are 24 GPS satellites (21 active satellites and 3 orbital spares) orbiting the earth.
- Orbits are roughly circular
- Altitude of 20,163 km above the earth's surface

http://www.youtube.com/watch?v=IPh6R30UJrg&feature=related



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS Satellites

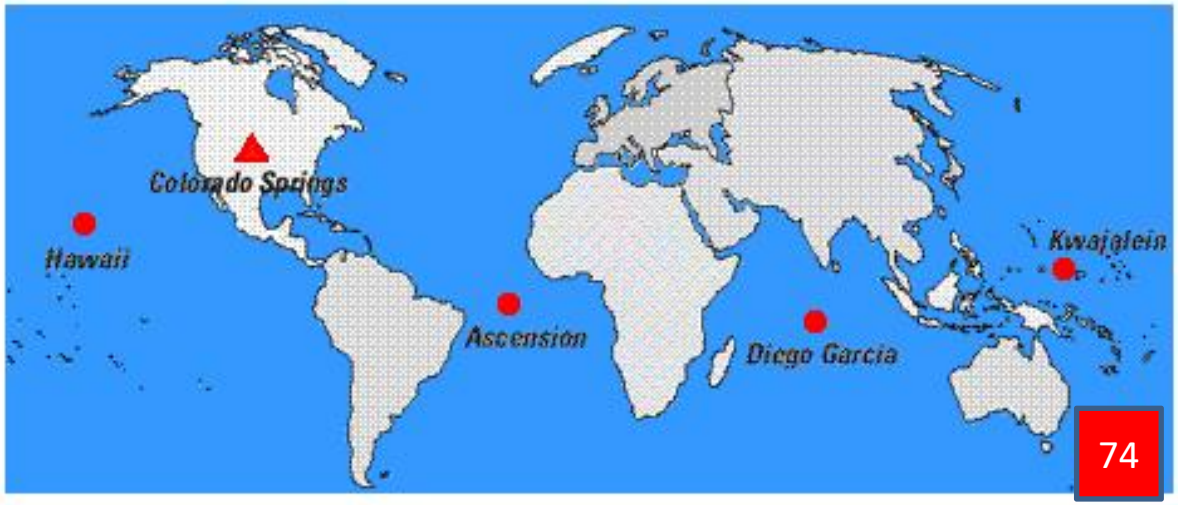
- GPS satellites **transmit precisely timed GPS signals and the data stream for positioning**.
- Timing signals are referenced to **atomic clocks**,
- A 24-satellite configuration provides adequate satellite coverage to perform 3 position fixing.
- Each satellite follows a unique path over the ground.
- Every satellite operates between 55 degrees North and 55 degrees south.
- These satellites move in their orbits, the satellites in view at each location changes with time.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Control Segment

- 5 monitoring stations are located throughout the world provide continuous surveillance and upload information to the satellite
  - Hawaii,
  - Colorado Springs,
  - Ascension Island,
  - Diego Garcia
  - Kwajalein Island

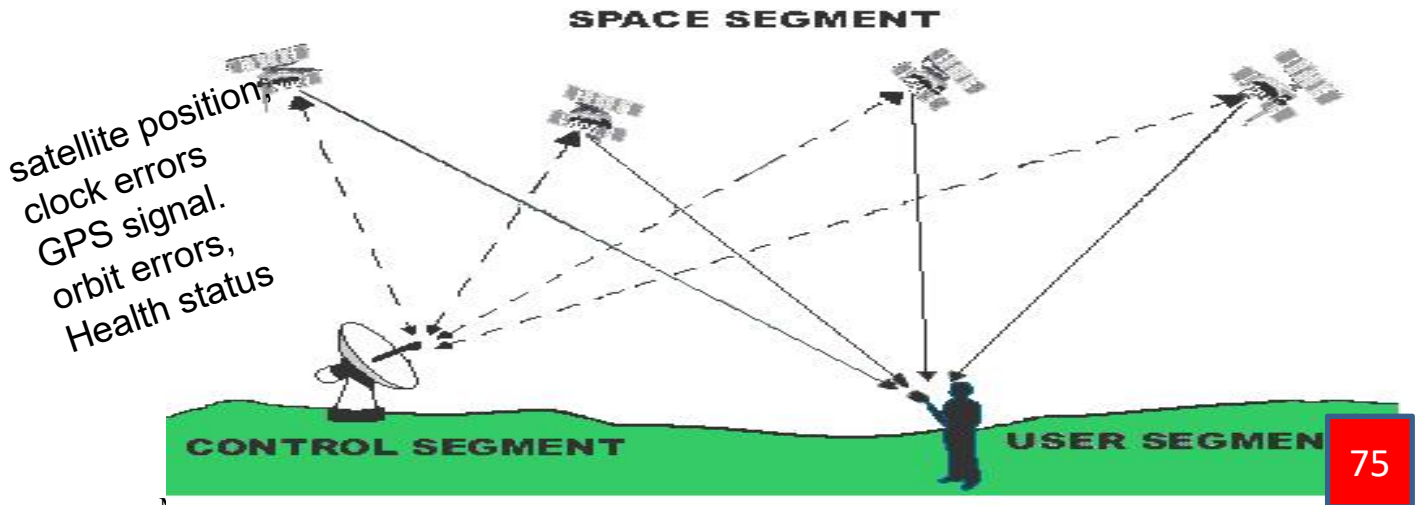


Courtesy: Dr. Wilton Fok

Most n

Control Segment

- The objective of the GPS control segment is to:
  - **Maintain each of the satellites in its proper orbit** through infrequent, small commanded maneuvers,
  - **Make corrections and adjustments** to the satellite clocks and payload as needed,
  - **Track the GPS satellites** and generate and upload navigation data to each of the GPS satellites, and
  - **Command major relocations** in the event of **satellite failure** to minimize the impact.



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Control Segment

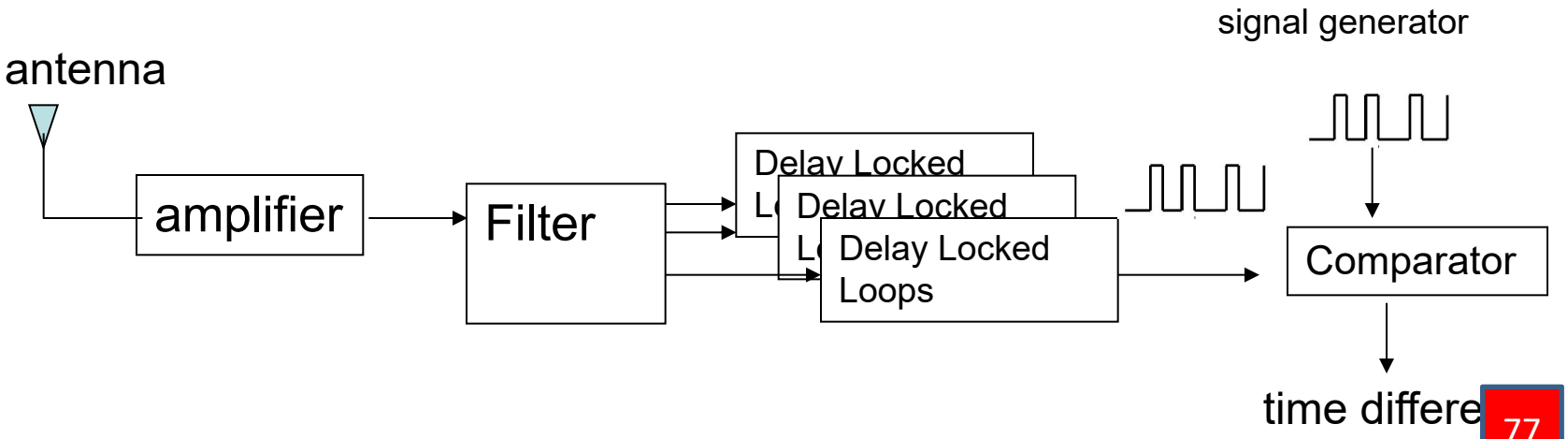
- **Navigation data packages** are then prepared for uploading to the satellites via the ground antenna stations for storage and use on the satellites.
- Uploads rate: 1-3 times per day
- Uploaded data can be used for up to 14 days
  - Provide a degree of autonomy should there be difficulties in uploading data for an extended period of time.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS User (Receiver)

- Antenna receives the GPS signals and amplifies them
- A filtering eliminates signals or noise **from adjacent freq. bands.**
- The signal is then fed to sets of delay locked loops where **multiple satellites can be tracked simultaneously**.
- The carrier phases and navigation data is then **estimated**.
- A signal generator replicating the signal produced by the satellites is used to **determine the time difference** between when the signal was transmitted by the satellite and received by the user.

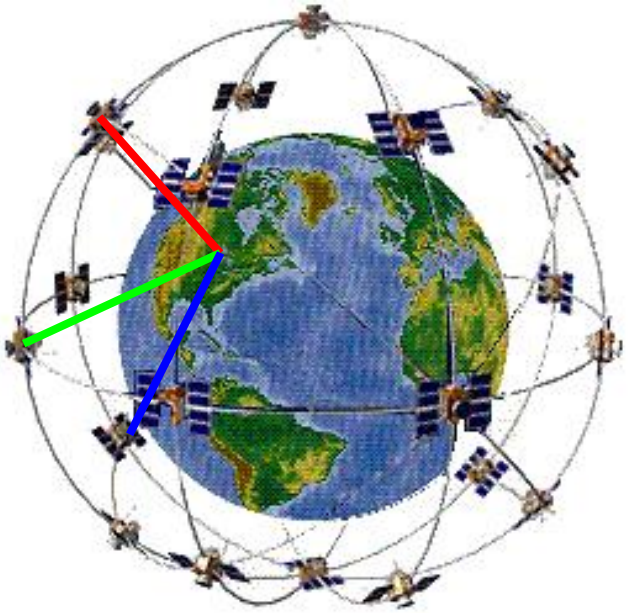


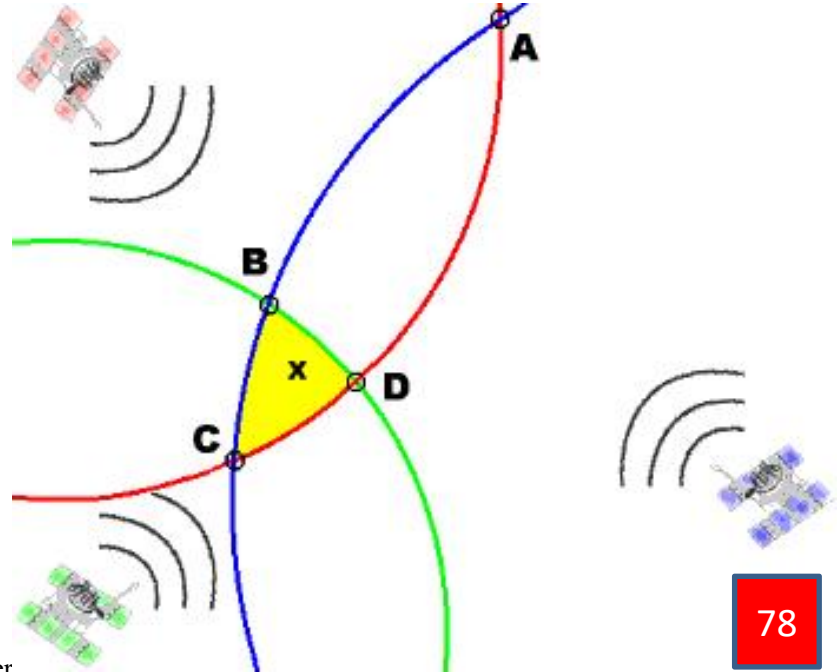
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

GPS User (Receiver)

- This information is then processed with other sensory data (if available) to produce a position and velocity output.
- The coordinates are then converted by the appropriate geodetic transformation to the local coordinate set.



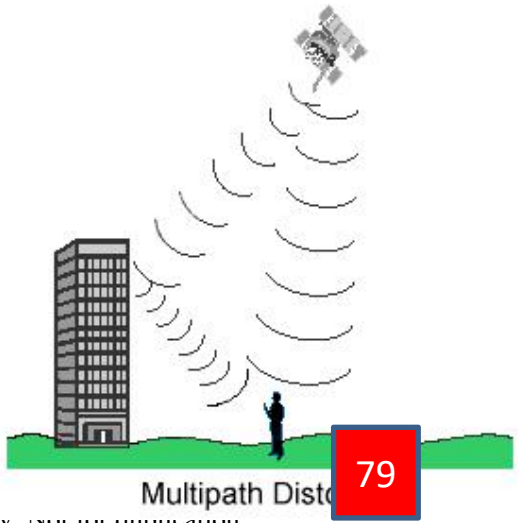


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Multipath Error

- Multipath is the effect of the same satellite signal reaching the GPS antenna more than once.
- The first signal is **direct path** from the satellite.
- The multipath signals are **reflected by building, water or ground surfaces**



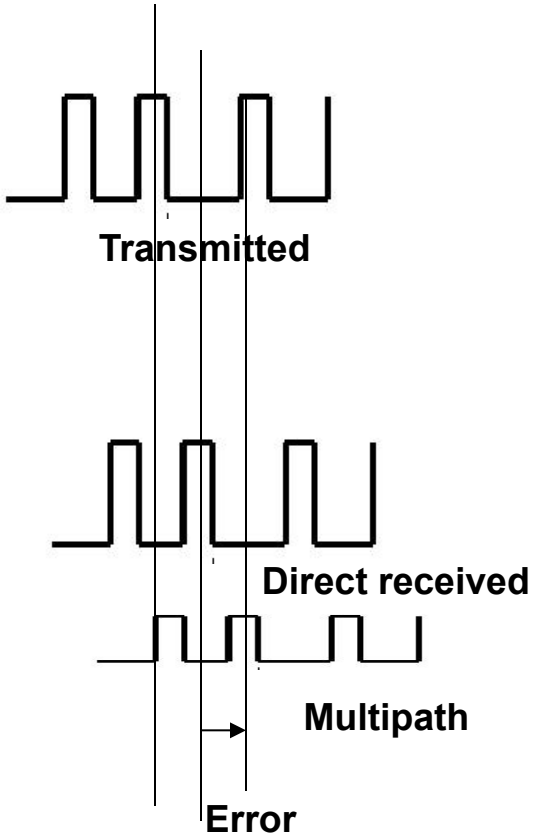
Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Multipath Error

- The reflected signal arrives slightly later than the original, resulting in incorrect range calculations.

- Multipath errors can be addressed by
- **software algorithms**
  - reject the multipath signal and lock on to direct signal.
- **special antennas with rounded metallic**
  - reflect the unwanted signals and shielding for interference



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

BeiDou Navigation Satellite System

- **BDS** is a Chinese satellite navigation system.

- **BeiDou-1** consists of 4 satellites (3 active + 1 spare)

- **BeiDou-2** consists of 35 satellites
- **BeiDou-3**, consists of 30 satellites.
- The constellation was declared "complete" on 23 June 2020.



[1]Update on the BeiDou Satellite Navigation System Archived 23 October 2018 at the Wayback Machine. 12th ICG Meeting. Jia-Qing Ma, China Satellite Navigation Office.

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

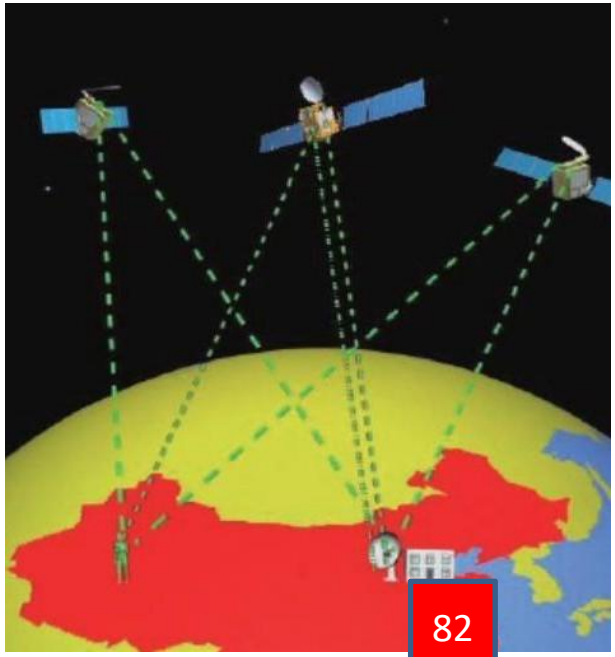
BDS Position calculation

- Steps to calculate a position
  1. A signal is transmitted skyward by a remote terminal.
  2. Each of the geostationary satellites receive the signal.
  3. Each satellite sends the accurate time of when each received the signal to a ground station.

4. Ground station calculates the longitude and latitude of the remote terminal, and determines the altitude from a relief map.

5. The ground station sends the remote terminal's 3D position to the satellites.

6. The satellites broadcast the calculated position to the remote terminal.

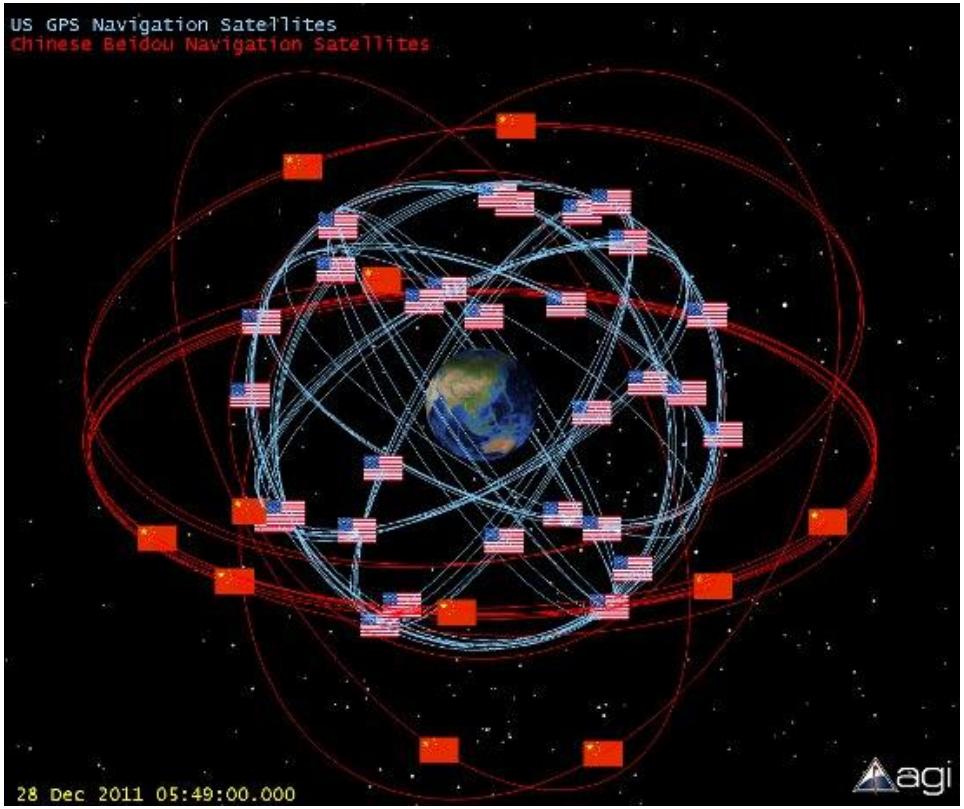


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

BeiDou Satellite System (BeiDou-2)

- Consist of 35 satellites for the complete coverage of the globe.
  - 5 geostationary orbit satellites for backward compatibility with BeiDou-1
  - 30 non-geostationary satellites (medium earth orbit)

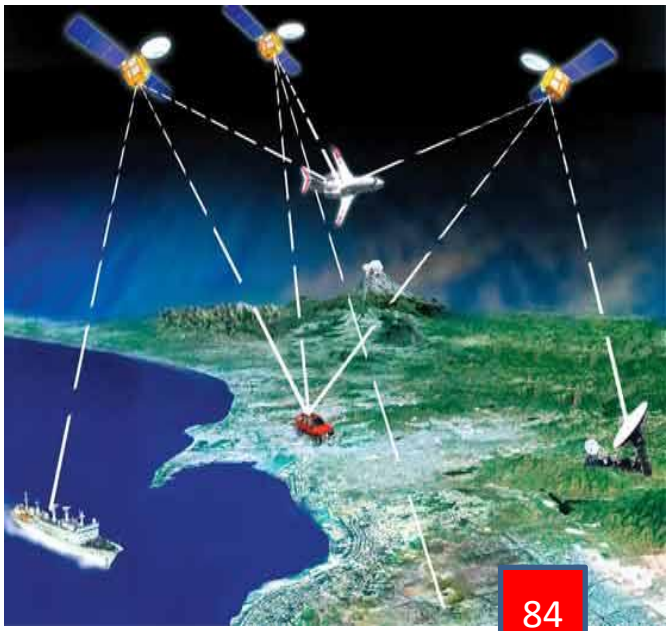


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

BeiDou-2

- 2 levels of service
  - **free service to civilians**
    - 10m location-tracking accuracy
    - Synchronize clocks with an accuracy of 10 ns
  - **licensed service t**o Chinese government and military users



Courtesy: Dr. Wilton Fok

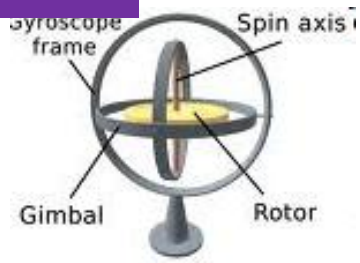
Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Features of Beidou

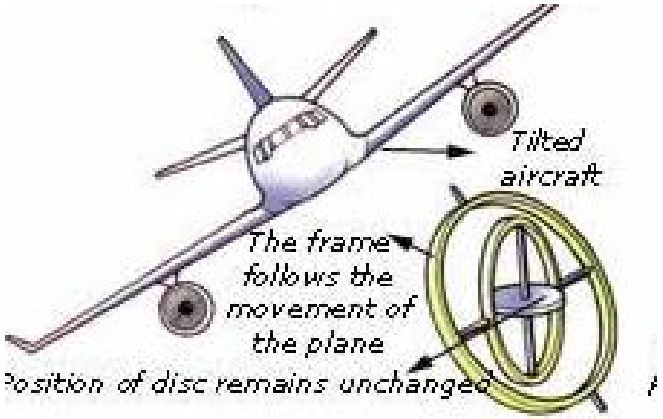
Example 3.5 Features of Beidou

Q: What are the features of Beidou?

- **1. Dual-Mode Functionality:**
- BeiDou offers both **global and regional services**, initially focusing on regional applications in China and gradually expanding to provide global coverage.
- **2. Short Message Communication:**
- Unlike most GNSS, BeiDou includes the ability for users to send short text messages, facilitating two-way communication between the satellite and users.
- **3. Interoperability:**
- BeiDou is designed to be interoperable with other GNSS systems, such as GPS, GLONASS, and Galileo, enhancing global navigation capabilities.
- **4. Advanced Anti-Jamming Capabilities:**
- The system incorporates sophisticated anti-jamming technology, improving its resilience against interference and enhancing reliability in challenging environments.
- **5. Ground Augmentation (quite unique)**
- BeiDou utilizes a ground-based augmentation system to enhance position accuracy and reliability. **By routing signals through ground stations, the system can perform corrections and adjustments before sending the signals to users**, improving overall service quality.



http://en.wikipedia.org/wiki/Gyroscope



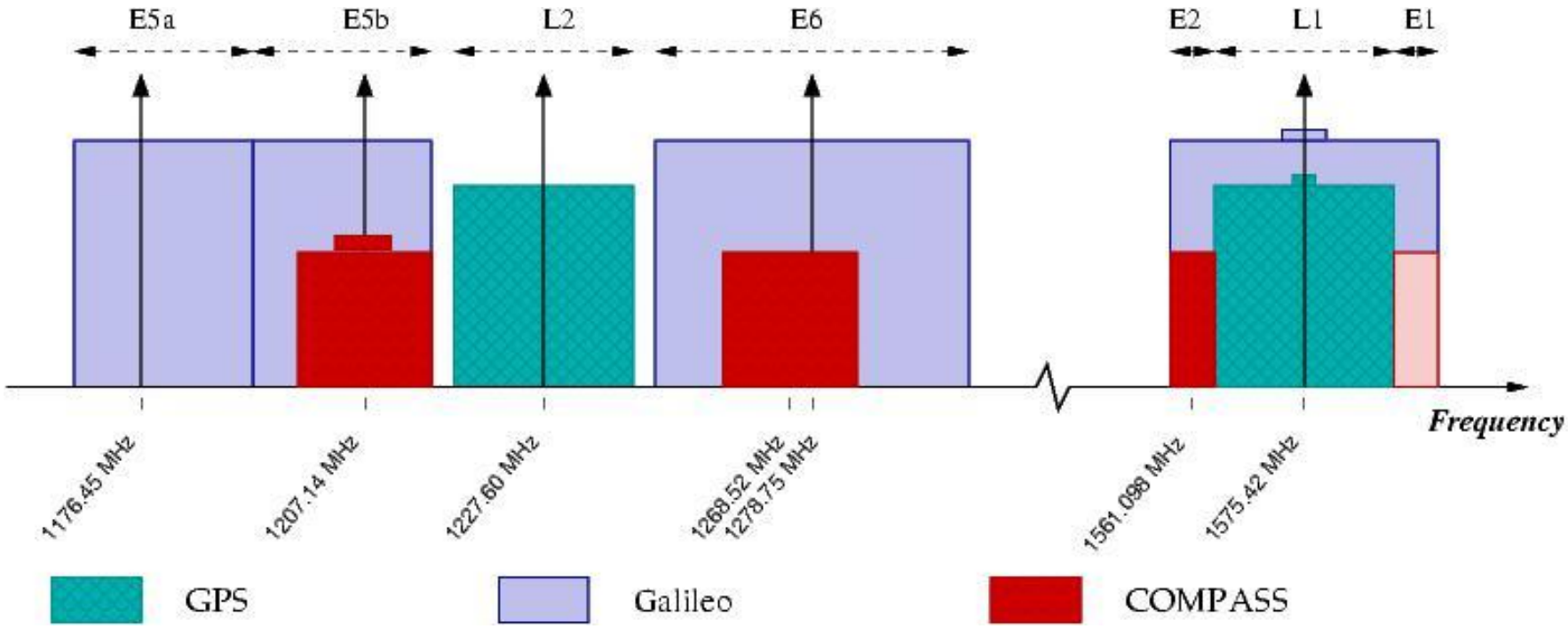


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

BDS Ranging signal

- Based on the CDMA principle
- Frequency allocation of GPS, Galileo, and Compass

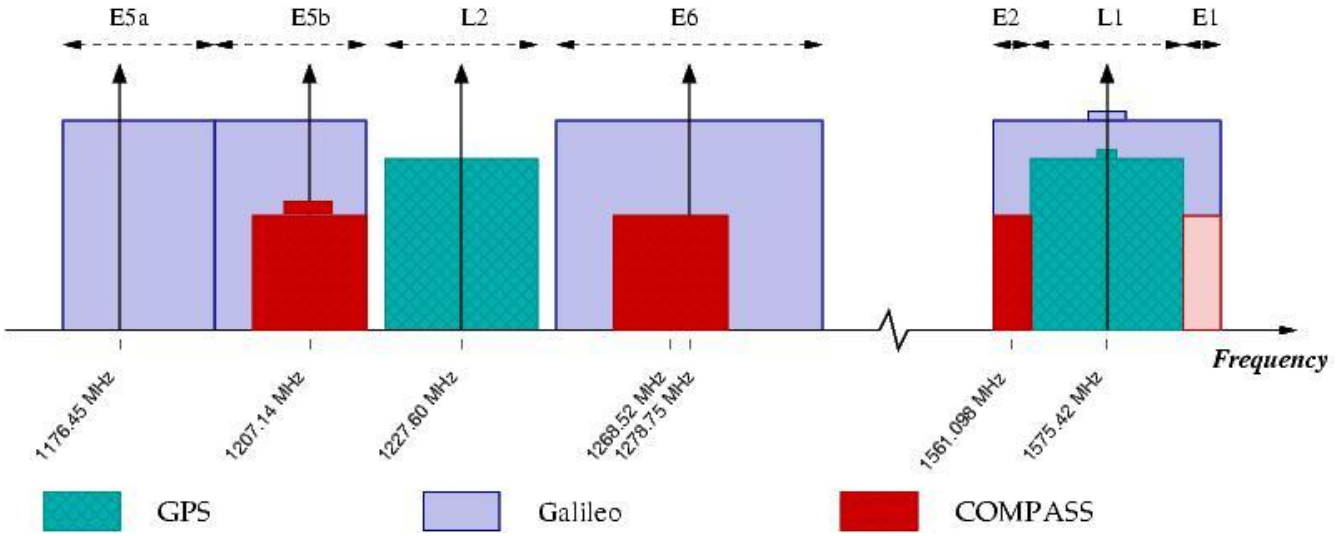


Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

BDS Ranging signal

- Frequencies for BDS are allocated in four bands: E1, E2, E5B, and E6 and overlap with Galileo.
- Pros:
  - Compatible receiver design
- Cons
  - Inter-system interference, especially within E1 and E2 bands (allocated for Galileo's publicly regulated service)



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

More about GPS

- GPS: Theory, Practice and Applications Frederic G. Snider, R.P.G.
  - http://www.pdhonline.org/courses/l116/l116content.pdf

Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Appendices for GSM

TE (Terminal Equipment): The device used by the user to access the mobile network, such as a mobile phone or modem.

R (Radio): Refers to the radio interface, which includes the transmission and reception of radio signals between the mobile device and the base station.

MT/TE (Mobile Termination/Terminal Equipment): The process of terminating a call to a mobile phone (MT) or the device itself (TE) involved in that termination.

Um: The air interface between the Mobile Station (MS) and the Base Station (BS), facilitating communication over radio frequencies.

A-bis: The interface between the Base Station Controller (BSC) and the Base Transceiver Station (BTS), managing the connection and communication between these components.

A: The interface between the Mobile Switching Center (MSC) and the Home Location Register (HLR), responsible for handling call routing and subscriber data.

Not Examined



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.

Appendices for GSM

MSC (Mobile Switching Center): The central component that manages call routing and mobile communication within the network. It connects calls between mobile users and between mobile and fixed networks.

VLR (Visitor Location Register): A temporary database that stores information about subscribers currently within the jurisdiction of the MSC. It helps manage user location and service information.

SGSN (Serving GPRS Support Node): A node that manages data sessions for mobile devices using the General Packet Radio Service (GPRS). It handles packet routing and mobility management.

H/E (Handover/Equipment): Refers to the process of transferring an ongoing call or data session from one cell or base station to another (handover) and the equipment involved in this process.

Gr/Gs: Interfaces used for communication between the MSC and the SGSN. Gr is between the MSC and the SGSN for GPRS services, while Gs allows for signaling and coordination between the circuit-switched and packet-switched domains.

Gb: The interface between the SGSN and the Radio Network Controller (RNC) or Base Station Controller (BSC), responsible for data transfer in packet-switched networks.

Gn: The interface between different SGSNs, allowing for user data and signaling transfer across GPRS networks.

Gi: The interface between the GPRS network and external packet data networks, providing connectivity to the internet or other data services.



Courtesy: Dr. Wilton Fok

Most materials are from the text & reference books. For use in lectures only. Not for duplication.